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Molecular dynamics of wettability and condensation on nanostructured surfaces: Fundamentals and hybrid wetting

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ABSTRACT

In this review article, molecular dynamics simulation is utilized to study wettability and condensation phenomena on nanostructured surfaces with combined wettability characteristics (hydrophilic and hydrophobic). Nanostructured surfaces, through the combination of various wettability properties, have demonstrated significant improvements in heat transfer and the stability of droplet condensation processes. Specifically, surfaces with hybrid wettability have shown to enhance condensation efficiency by up to 10 times compared to traditional hydrophilic surfaces, as indicated by simulation studies. By optimizing roughness and geometric variations, these surfaces reduce thermal resistance by approximately 20–30 %, enabling more efficient thermal management. Furthermore, molecular dynamics simulations have revealed critical parameters influencing droplet growth, such as the impact of hydrophilic-to-hydrophobic ratios, where higher hydrophilic content accelerates nucleation by 40–50 %. This research also addresses existing challenges, including surface stability under varying environmental conditions and the need for experimental data to validate simulation models. Emphasizing these findings, the study highlights the importance of combined wettability technologies in applications such as nanomedicine, microfluidics, and energy storage.

1. Introduction

The study of condensation and wettability has garnered significant attention due to its important role in thermal management, energy systems, and various industrial applications. Understanding the fundamental mechanisms of condensation and the influence of surface properties is essential for optimizing these processes. This introduction is structured to provide an overview of key concepts, beginning with an examination of the mechanisms and importance of condensation. It then explores the impact of wettability and geometric parameters, the role of molecular dynamics (MD) simulations in advancing our knowledge, and finally, discusses the innovations and methodologies employed in this review.

1.1. Condensation: mechanisms and importance

Condensation is one of the critical phenomena in thermal sciences and mass transfer, playing an essential role in many industrial processes, including energy production, cooling systems, and air conditioning. This phenomenon occurs when vapor converts into liquid, usually due to a drop in temperature or an increase in vapor pressure. Condensation can occur either as filmwise condensation or dropwise condensation, each with its specific characteristics and applications [1]. In filmwise condensation, a thin layer of liquid forms on the cold surface, gradually flowing downward. This type of condensation typically occurs on smooth, hydrophilic surfaces and, due to the continuous liquid layer, results in less efficient heat transfer [2]. In contrast, dropwise condensation happens when the surface is hydrophobic, causing the vapor to condense into small droplets rather than forming a continuous film. These droplets then coalesce and roll off the surface, leading to a significant increase in heat transfer rates. Since dropwise condensation can greatly enhance heat transfer efficiency, designing surfaces that promote this type of condensation has been a key focus in recent research [3]. In this regard, surface wettability and geometric parameters play a crucial

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role in determining the type and efficiency of condensation. Surfaces with appropriate roughness or specific surface patterns can exhibit both hydrophilic and hydrophobic properties simultaneously, known as combined wettability, which enhances condensation efficiency. Nanostructured condensation phenomena have also gained particular attention, as at the nanoscale, surface effects and the geometric parameters of nanochannels significantly influence condensation behavior (Fig. 1) [4].

Although significant progress has been made in understanding the mechanisms of condensation, existing models still face challenges that limit their accuracy and applicability. Many studies focus primarily on ideal behaviors, such as entirely hydrophilic or hydrophobic surfaces, while neglecting the complex behaviors observed on surfaces with a combination of hydrophilic and hydrophobic properties (hybrid surfaces). For example, the interaction between hydrophilic and hydrophobic regions on hybrid surfaces, especially concerning droplet coalescence and movement, has not been sufficiently investigated within experimental and theoretical frameworks.

Another critical gap lies in the limited understanding of heat and mass transfer phenomena at the nanoscale during the condensation process. Molecular dynamics (MD) simulation models, despite their importance, often cannot simulate large-scale or long-term condensation processes due to computational limitations. Furthermore, the lack of standardized methods for experimentally validating MD simulations has led to discrepancies between theoretical predictions and actual observations.

Additionally, challenges such as surface degradation under various environmental conditions and the impact of nanochannel geometry on condensation behavior remain inadequately explored. For instance, optimizing the balance between surface roughness and hydrophilicity for efficient droplet growth and removal remains an open question. Bridging these gaps requires integrating multiscale simulation methods with experimental studies, as well as advancements in fabrication technologies to develop robust and cost-effective nanostructured surfaces.

1.2. Wettability and influence of geometric parameters

Wettability and geometric parameters are among the key factors that determine condensation behavior in nanochannels. At the nanoscale, surface effects become significantly more critical than at larger scales, so even slight changes in wettability and geometric features can lead to substantial differences in nanochannel performance [6]. Due to their small size, nanochannels have an extremely high surface-to-volume ratio, which makes wettability and surface parameters crucial in heat and mass transfer processes [7]. Wettability, which refers to the interaction between a liquid and a solid surface, directly influences whether the liquid forms droplets on the surface or spreads out as a film [8]. Hydrophilic surfaces tend to spread liquids into films, while hydrophobic surfaces encourage droplet formation. In nanochannels, these properties directly impact the condensation process. Surfaces with combined wettability, meaning a mix of hydrophilic and hydrophobic areas, can create optimal conditions for condensation, enhancing heat transfer efficiency.

The geometric parameters of nanochannels are also highly important. The shape, size, and surface roughness of the channels can influence fluid flow and the condensation process. Nanochannels with varying cross-sectional areas can direct flow in such a way that optimizes temperature and pressure distribution within the channel. Additionally, nanochannels with rough or micro−/nanostructured surfaces can facilitate condensation by increasing surface contact and improving wettability [9]. Surface roughness plays a particularly critical role in condensation processes in nanochannels. It can increase the contact area between vapor and solid surfaces, thereby accelerating the condensation rate. In some cases, surface roughness can even alter wettability behavior, resulting in significant improvements in heat transfer. Surfaces with controlled roughness can help reduce fluid flow resistance and optimize heat transfer processes. Hybrid or combined wettability, which involves creating both hydrophilic and hydrophobic regions on a single surface, has emerged as an innovative approach in optimizing nanochannels [3]. This combination can guide fluid flow and ensure a more uniform temperature distribution along the nanochannel. Moreover, combined wettability can encourage droplet formation in specific areas of the surface, thereby dramatically increasing heat transfer rates. Theoretical modeling and simulations play an important role in understanding the impact of wettability and geometric parameters on the behavior of nanochannels [10]. Various applications of wettability are depicted in Fig. 2.

1.3. Role of molecular dynamics in understanding wetting

MD is one of the advanced computational methods used to simulate physical and chemical systems at the nanoscale. This method allows us to observe the behavior and interactions between atoms and molecules over time with high precision [11–13]. By modeling the motion of each atom or molecule based on Newtonian mechanics, MD enables the detailed study of processes such as wettability, condensation, heat

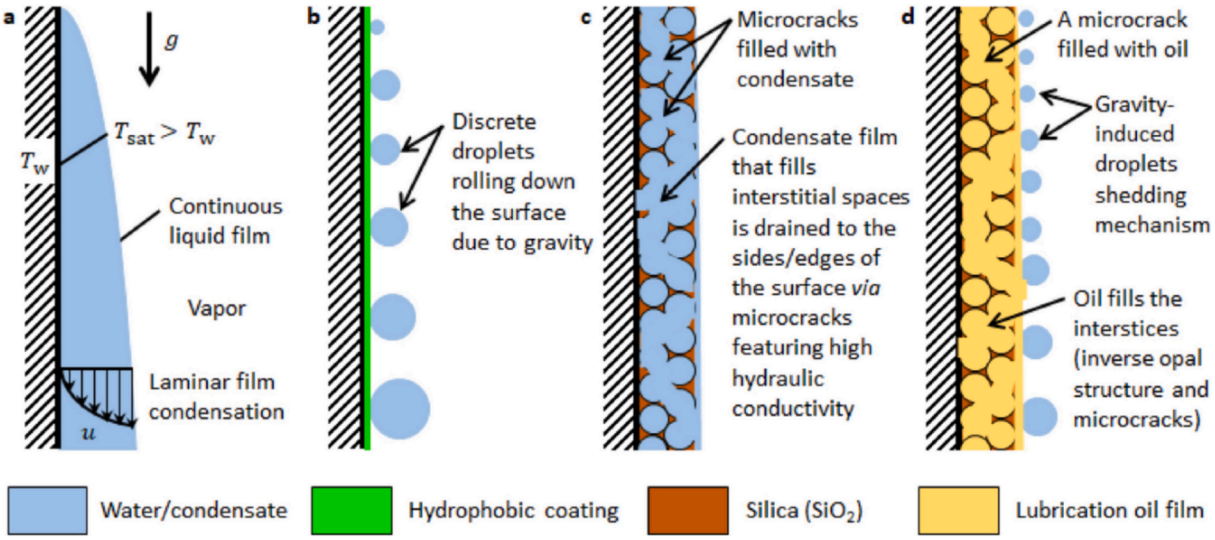

Fig. 1. Schematic illustrations of different condensation states [5].

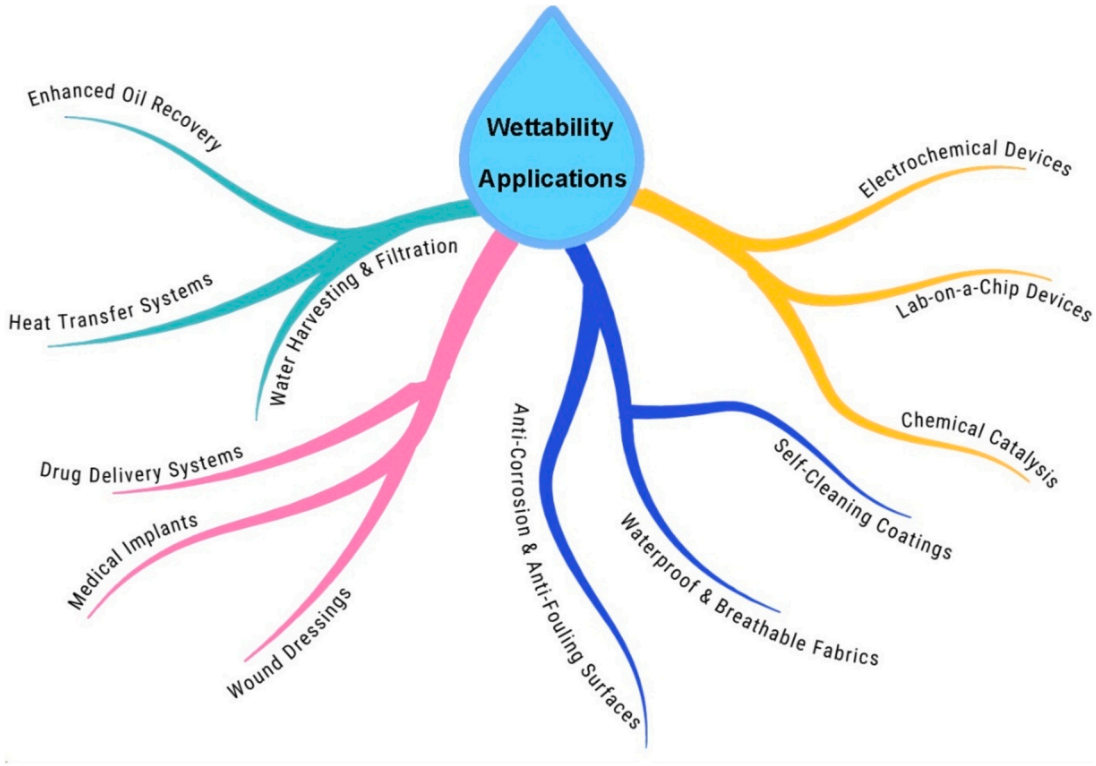

Fig. 2. Various applications of wettability.

transfer, and even chemical reactions. One of the key advantages of MD at the nanoscale is its ability to analyze phenomena that are not easily observable in conventional experiments. For example, in condensation and wetting processes, the surface characteristics of nanomaterials, such as contact angle and nanostructures, have a significant impact on fluid behavior. Using MD, surface modifications can be accurately simulated, and the effects of these changes on physical processes can be investigated [14,15]. In MD simulations, intermolecular forces play a crucial role in determining the system's behavior. Forces such as van der Waals interactions, hydrogen bonds, and electrostatic forces influence how molecules move and interact with one another. These forces are particularly important at the nanoscale, where their influence on material behavior is much greater than at larger scales [16,17]. Fig. 3 presents various methods of molecular dynamics. Different MD simulation approaches are available depending on the system type and simulation goals. One common method in MD is the use of the Lennard-Jones potential, which is often applied to model interactions between simple atoms and molecules.

This potential allows us to understand the attractive and repulsive forces between molecules and examine their effects on the overall system behavior. In MD, time and distances are modeled discretely. Simulations are carried out in very small-time intervals (femtoseconds) to accurately observe changes in the motion and energy of atoms. This high temporal precision enables the simulation and analysis of rapid and transient molecular phenomena, such as condensation nucleation or droplet formation [19]. Another important application of MD is studying the wettability of various surfaces. By simulating the interactions between water molecules and different surfaces, one can precisely analyze wettability behavior and the formation of dropwise or filmwise condensation. This is particularly significant in industries such as

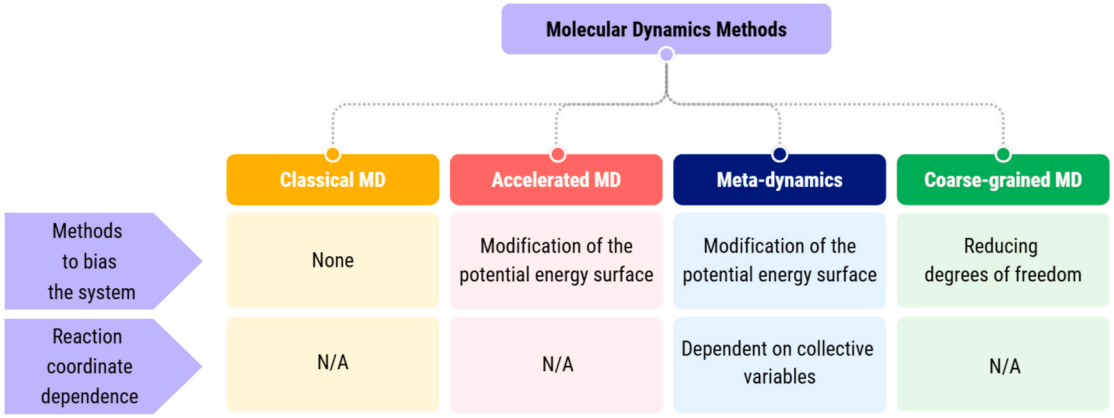

Fig. 3. Illustration of different MD simulation methods, highlighting their application areas (e.g., Lennard-Jones potential for van der Waals forces) and time resolution capabilities. These methods enable precise modeling of nanoscale phenomena such as wettability, condensation, and droplet behavior on engineered surfaces [18].

electronics cooling, hydrophobic and hydrophilic surface design, and cooling systems. Due to its high accuracy and ability to investigate nanoscale phenomena, MD has become a key tool in the development of innovative technologies. Since conventional experimental methods cannot directly access the nanoscale, MD allows researchers to theoretically and computationally study phenomena and design innovative systems in various fields [20,21].

1.4. Novelty and methodology

This article reviews previous studies on combined wettability (hydrophilic and hydrophobic) and its impact on improving condensation and heat transfer processes. A key novelty of this study is the examination of works that employ MD simulations to explore the behavior of condensate droplets and wettability at the nanoscale. MD simulations provide deep insights into molecular forces and surface behavior in nanostructures, demonstrating how changes in surface geometry and roughness can enhance condensation stability and reduce thermal resistance. Moreover, this article specifically reviews surfaces with combined wettability properties and examines their application in improving system efficiency. In addition, the paper covers studies on the geometric parameters of nanochannels and their effects on the condensation process.

2. Nanostructured surfaces

Carbon-based nanomaterials, such as carbon nanotubes and graphene, are known for their extraordinary strength, electrical conductivity, and thermal properties. Fig. 4. categorizes various types of nanostructured materials based on their unique composition and structural properties, each with specific applications in scientific and industrial fields. These materials are widely used in electronics, energy storage, and composite materials. Graphene, in particular, is ideal for applications such as flexible electronics and energy storage systems like batteries and supercapacitors due to its flexibility and high conductivity. Metal nanoparticles, including those made of gold, silver, and platinum, are used in catalysts, biosensors, medical diagnostics, and as antimicrobial agents due to their high surface-to-volume ratio and unique optical properties. These nanoparticles play a critical role in nanomedicine and environmental sensing technologies. Liposomes and polymer nanoparticles are particularly important in the pharmaceutical and medical industries. Liposomes, composed of lipid vesicles, and polymer nanoparticles are primarily used for targeted drug delivery. These nanostructures enable controlled therapeutic release, improving drug efficacy and reducing side effects.

Fullerenes, dendrimers, and quantum dots are specialized nanostructures with unique chemical and optical properties. Fullerenes, with their spherical carbon arrangement, are used in photovoltaics and as antioxidants, while dendrimers, due to their branched structures, are well-suited for drug delivery and diagnostics. Quantum dots, known for their size-dependent optical properties, are widely used in display technologies, bio-imaging, and as fluorescent markers. Other nanostructures, such as nanocrystals and emerging types, offer additional functionalities, such as enhanced catalytic properties and quantum effects, making them valuable in fields like optoelectronics, sensors, and quantum computing. In summary, nanostructured materials possess diverse and tunable properties, enabling their use in a wide range of applications, from medicine and energy to electronics and material science. Their ability to alter physical, chemical, and optical properties at the nanoscale has made them a crucial part of modern technological advancements. Fig. 4 shows various types of nanostructured materials.

Nanostructured surfaces refer to surfaces engineered at the nanometer scale (typically between 1 and 100 nm) to obtain specific physical, chemical, or mechanical properties [23]. These surfaces exhibit unique behaviors, different from smooth and ordinary surfaces, due to their increased surface area and precise control over surface characteristics. By modifying surface roughness or creating nanoscale patterns, interactions with liquids, gases, or other materials can be controlled, making them highly useful for various technological applications [24,25].

One key feature of nanostructured surfaces is the ability to adjust wettability, which can enhance either hydrophobicity (water repellence) or hydrophilicity (water attraction). For example, by creating specific nanoscale patterns, a surface can become superhydrophobic, allowing water to form droplets that roll off the surface, like the lotus leaf effect. On the other hand, these surfaces can also be made superhydrophilic, encouraging liquids to spread evenly, which is especially beneficial in medical devices and water filtration.

Nanostructured surfaces also offer improved optical, mechanical, and thermal properties. In optics, they can reduce light reflection, increase light absorption in solar panels, or enhance lens clarity [26,27]. Mechanically, these surfaces can be harder, more resistant to wear, or even more flexible. In thermal applications, nanostructured surfaces are used to improve heat transfer processes such as condensation and evaporation in cooling systems, dramatically increasing efficiency. These surfaces have widespread applications, including self-cleaning surfaces where water droplets remove contaminants, and antibacterial coatings that prevent biofilm buildup in medical and industrial

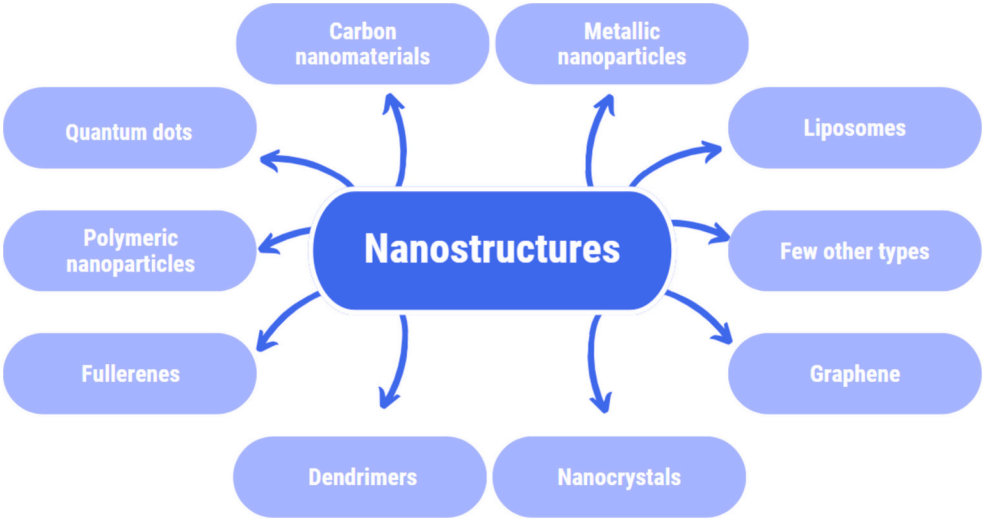

Fig. 4. Various types of nanostructured materials [22].

environments [28,29]. They are also employed in heat transfer systems, water collection technologies, energy storage devices like batteries and solar cells, and in sensitive sensors for detecting chemicals or biological agents.

Despite their many advantages, the broad use of nanostructured surfaces faces challenges such as high manufacturing costs and long-term durability. While these surfaces have great potential, particularly in energy and healthcare sectors, large-scale production remains expensive. Researchers are also focusing on improving the durability of these surfaces, especially in harsh environments, and ensuring their environmental sustainability so that nanomaterials can be developed for industrial use in a sustainable manner. These images illustrate various techniques for creating nanostructured surfaces with precise geometric control, which are essential for applications such as sensors, catalysts, optical devices, and thermal management systems [29–31]. Fig. 5 shows an SEM gallery of nanostructured surfaces using nanosphere lithography.

3. Fundamentals of wettability in condensation

3.1. Definition and measurement of wettability

Wettability is a key surface property that describes the interaction between a liquid and a solid surface, and it is important in many industrial and biological processes. Wettability refers to how a liquid behaves on a solid surface: if the liquid spreads evenly on the surface, the surface is considered hydrophilic, and if it forms droplets, the surface is considered hydrophobic [33]. The degree of wettability of a surface is determined by the contact angle between the liquid and the solid surface. The contact angle, as a quantitative measure of wettability, is the angle formed by the tangent to the liquid droplet at the point where it meets the solid surface. Surfaces with a contact angle of less than 90 degrees are considered hydrophilic, while surfaces with a contact angle greater than 90 degrees are considered hydrophobic. This measure effectively indicates the tendency of the liquid to either spread or bead

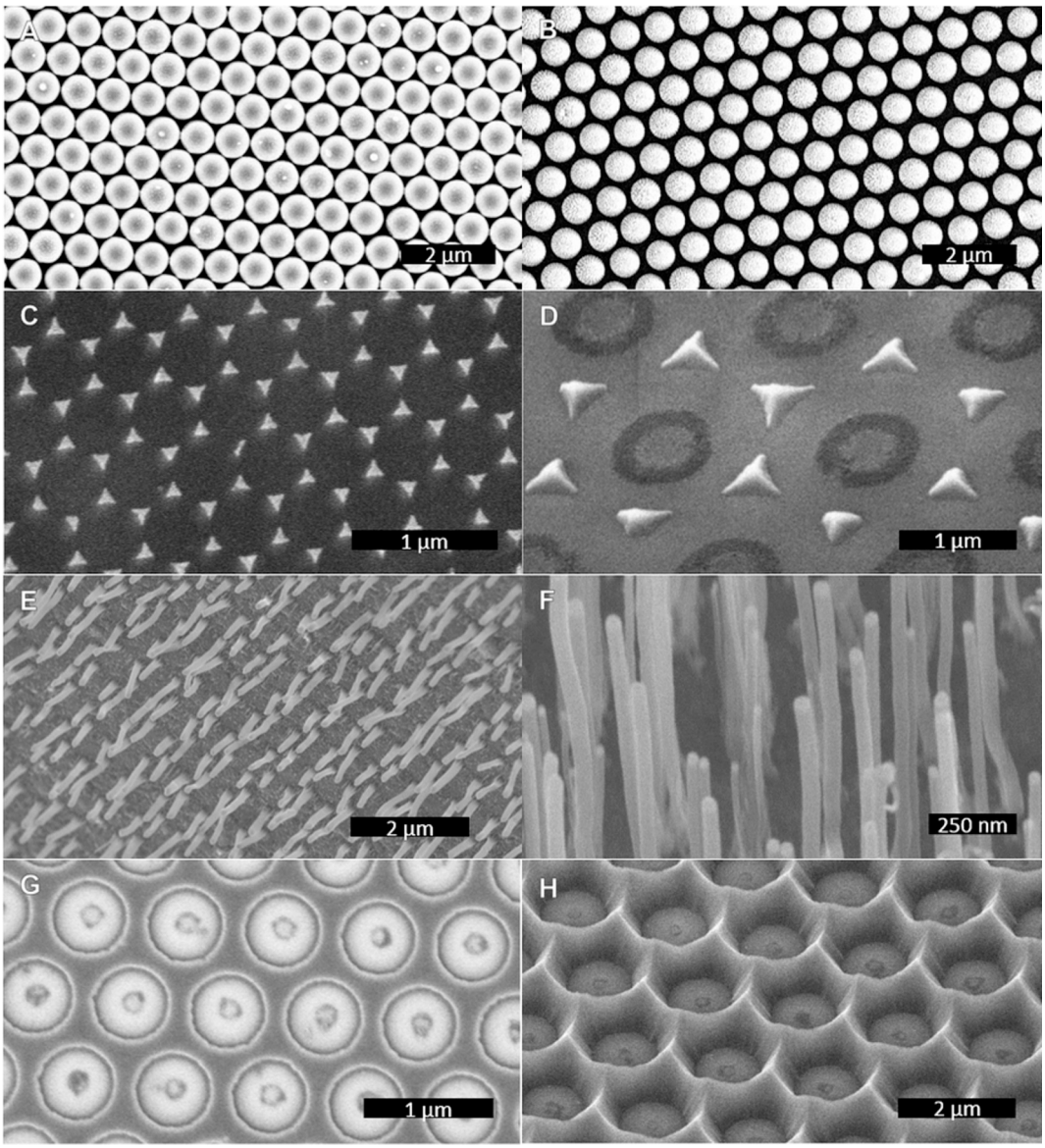

Fig. 5. Scanning Electron Microscope (SEM) images showcasing nanostructured surfaces created via nanosphere lithography. (A) Hexagonally close-packed polystyrene spheres (756 nm diameter). (B) Plasma-etched spheres reduced to 674 nm. (C) Arrays of 20-nm-thick triangular gold islands from 471-nm particles. (D) Pyramidal SiO2 islands (600 nm height) at 45° angle. Additional examples include perforated gold films (50 nm thick, 568 nm holes) and multi-walled carbon nanotube (MWCNT) structures. These images highlight the precision achievable with nanosphere lithography for functional surface design [32]. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

up on the surface and is widely used in various studies [34,35]. Fig. 6 shows schematic diagrams of conceptual models of surface wettability and contact angle hysteresis.

One of the most common methods for measuring contact angle is the Sessile Drop Method [37]. In this method, a small droplet of liquid is placed on a solid surface, and an image is captured using precise cameras. The contact angle is then calculated using image analysis software. Due to its simplicity and high accuracy, this method is widely used in research and industrial applications. Other methods, such as surface tension measurement, are also used to investigate wettability [38]. In this method, surface forces between the liquid molecules and the solid surface are evaluated. The data obtained from these methods can provide a deeper understanding of the molecular mechanisms behind wettability and aid in optimizing surfaces for specific applications. Wettability hysteresis is another important feature when examining surface behavior, representing the difference between the advancing contact angle and the receding contact angle [39]. This characteristic reflects the degree of liquid adhesion to the surface and the surface's resistance to liquid movement [40]. Wettability hysteresis is typically observed on surfaces with high roughness or heterogeneity and can affect the performance of surfaces in practical applications. Wettability on nanostructured surfaces is particularly important in industrial applications such as designing micro- and nanochannels, waterproof coatings, and self-cleaning surfaces [41]. In these applications, the ability to control wettability can improve system performance and efficiency. For instance, hydrophobic nanostructured surfaces can be effective in preventing water droplet formation in heat transfer systems or wind turbines. Wettability also plays a vital role in biological processes, such as cellular interactions and biofilm formation. Controlling the wettability of biocompatible materials can enhance biomolecular interactions and reduce the risk of infections. Accurate measurement and a deeper understanding of wettability in this context can lead to the development of biocompatible materials and optimization of medical devices [42].

3.2. Impact of wettability on condensation efficiency

Wettability, as one of the key surface properties, plays a crucial role in the condensation process and directly affects its efficiency. Condensation is an essential heat transfer process used in many industrial systems, such as heat exchangers, cooling systems, and distillation units [43]. The type of surface wettability (hydrophilic or hydrophobic) largely determines whether condensation occurs as filmwise or dropwise, which significantly impacts the rate of heat transfer. In filmwise condensation, liquid spreads over the surface as a thin, continuous film (Fig. 7) [6,44]. This type of condensation typically occurs on hydrophilic surfaces. The film layer formed on the surface creates a higher thermal resistance, hindering optimal heat transfer from the surface to the liquid. As a result, thermal efficiency is reduced, making filmwise condensation undesirable in many industrial applications, especially where high heat transfer rates are needed. In contrast, dropwise condensation occurs on hydrophobic surfaces [45]. In this case, vapor condenses into small droplets on the surface, and instead of forming a continuous layer, the droplets combine and roll off the surface. This process results in a significant increase in the heat transfer rate because thermal resistance is much lower than in filmwise condensation. Studies have shown that dropwise condensation can improve thermal efficiency by up to 10 times compared to filmwise condensation [46].

Surface wettability can also impact the stability of dropwise condensation. Surfaces with combined wettability, meaning a mix of hydrophilic and hydrophobic areas, can enhance the stability of dropwise condensation [48]. This combination effectively prevents the

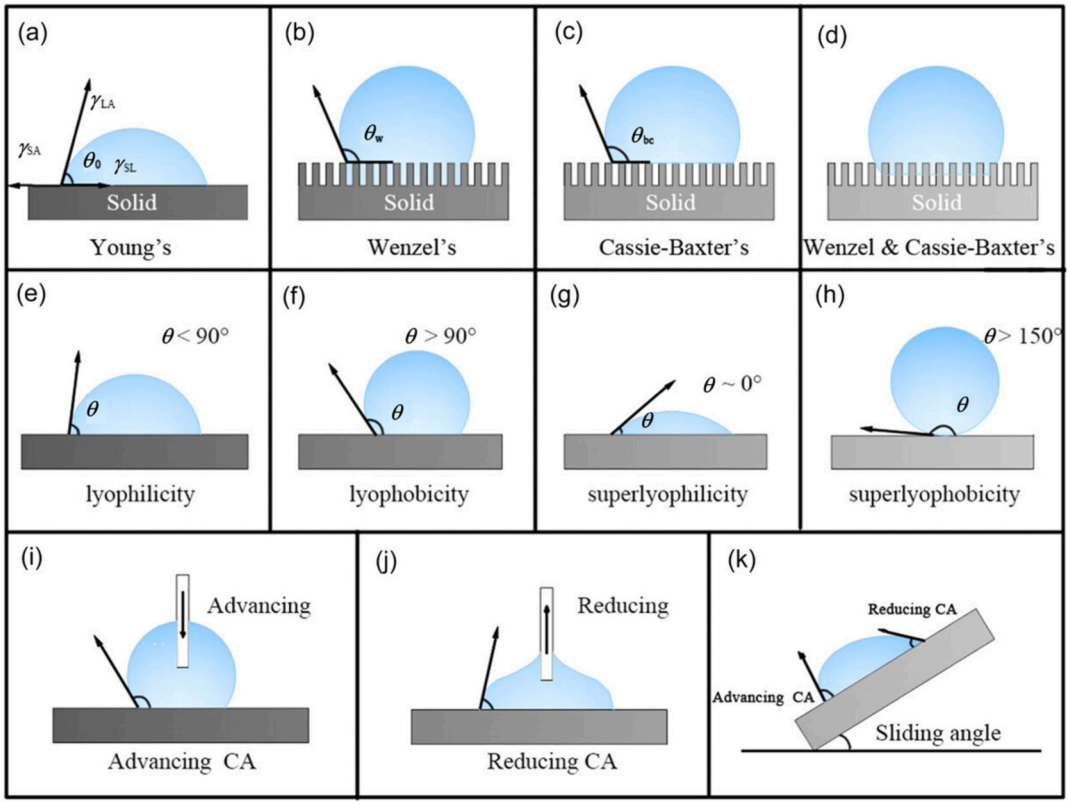

Fig. 6. Key models of wettability and the principles behind oil/water separation. (a–d) Illustrative diagrams of theoretical models depicting surface wettability and contact angle hysteresis: (a) Young's model, (b) the Wenzel model, (c) the Cassie–Baxter model, and (d) the combined Wenzel & Cassie–Baxter model. (e–h) Diagrams showing the behavior of drops on a solid surface in different states: (e) lyophilic, (f) lyophobic, (g) superlyophilic, and (h) superlyophobic. (i–k) Diagrams outlining the dynamic parameters of wettability: (i) advancing contact angle, (j) receding contact angle, and (k) sliding angle. [36].

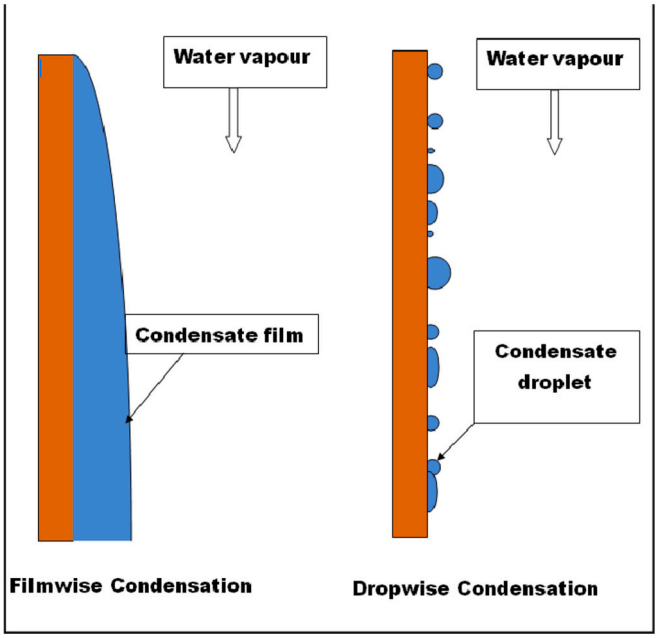

Fig. 7. Schematic of filmwise and dropwise condensation on a smooth surface [47].

formation of large droplets and the reformation of a liquid film, thereby maintaining a high heat transfer rate over an extended period. The type of material and its surface characteristics also significantly influence condensation behavior. Nanostructured and microstructured surfaces can dramatically improve condensation rates and thermal efficiency by altering surface roughness and optimizing wettability. These surfaces can also help reduce wettability hysteresis and increase droplet stability [49,50]. From an application perspective, controlling surface wettability can lead to improved performance in industrial systems. In heat exchangers, using hydrophobic or combined wettability surfaces can increase heat transfer rates and reduce system size. Additionally, in cooling and distillation systems, optimizing surface wettability can reduce energy consumption and enhance overall system efficiency [51].

4. Hybrid wetting: concept and mechanisms

4.1. Introduction to hybrid wetting

Hybrid wetting refers to a condition where a surface exhibits both hydrophilic (water-attracting) and hydrophobic (water-repelling) properties. This dual wettability allows for more complex liquid-surface interactions compared to surfaces that are purely hydrophilic or hydrophobic. The coexistence of these regions on a single surface enables controlled liquid behavior, which is essential for processes such as condensation, heat transfer, and droplet manipulation. The presence of hydrophilic regions attracts liquid, while hydrophobic regions repel it, creating a balance that governs how droplets form, move, and interact with the surface [52,53]. These behaviors can be exploited for advanced functionalities, such as self-cleaning surfaces and directional liquid transport [54,55].

4.2. Mechanisms of droplet behavior on hybrid surfaces

On hybrid wetting surfaces, the behavior of droplets is influenced by the spatial arrangement of hydrophilic and hydrophobic regions. Hydrophilic areas act as droplet collection points due to their stronger adhesive forces, whereas hydrophobic areas promote droplet movement or detachment. This combination of wetting characteristics can control droplet behavior, including spreading, coalescence, and movement, providing a versatile mechanism for precise liquid management. For example, droplet movement is governed by the balance of surface tension forces, and the transition from hydrophilic to hydrophobic regions can result in directional droplet transport [52,53].

In MD studies, Paul et al. [56] observed that hybrid surfaces with higher hydrophilic content facilitated faster nucleation and droplet growth. This highlights how the distribution of wetting properties affects liquid behavior at the nanoscale. In a related study, Zhang et al. [57] used a hybrid MD/VOF simulation approach to study droplet spreading on surfaces, focusing on three-phase contact line dynamics. Their multiscale method (Fig. 8) combines MD to provide boundary data for interfacial slip and contact angle, while VOF covers the entire domain. The model was validated against full MD simulations for water nanodroplets on platinum, showing the importance of accounting for slip, especially at the contact line. This method was extended to larger droplets (up to 100 nm), where full MD is computationally impractical, with slip effects decreasing as droplet size increases. (See Fig. 9.)

4.3. Role of surface roughness in hybrid wetting

Surface roughness plays a critical role in enhancing hybrid wetting behaviors. At both the micro- and nanoscale, roughness can amplify the effects of hydrophilic and hydrophobic regions. For example, hydrophilic areas with roughness can act as droplet pinning centers, which helps to stabilize droplet formation and prevents them from spreading uncontrollably. Conversely, hydrophobic areas with nano-roughness can enhance liquid repellency by preventing the droplet from fully wetting the surface [58,59].

Wagemann et al. [60] demonstrated this in their study of nanostructured hexagonal boron nitride surfaces, where the roughness and edge morphology of hexagonal boron nitride nanoribbons influenced wettability. Zigzag edges were found to be more hydrophilic, while armchair edges were more hydrophobic. This example highlights how the design of nanoscale surface structures, such as roughness and edge properties, can significantly alter the wetting behavior of hybrid surfaces.

In addition, Quéré [61] examined the impact of surface roughness on wetting properties and showed how increased surface roughness can shift a system from a wet to a non-wet state, depending on the combination of surface tension forces. This roughness-driven transition between different wetting states plays a key role in hybrid wetting mechanisms, affecting how liquid behaves across a heterogeneous surface.

4.4. Nucleation and growth in hybrid wetting

Hybrid wetting surfaces also have a pronounced impact on nucleation and droplet growth, particularly in condensation processes. In hybrid wetting, hydrophilic regions tend to act as nucleation sites, attracting liquid molecules and facilitating droplet growth, while hydrophobic regions limit the spreading of these droplets, allowing for controlled droplet sizes. This behavior is critical in achieving efficient dropwise condensation, where smaller droplets enhance heat transfer rates [58,59].

Gao et al. [62] investigated the process of water vapor condensation on nanostructured surfaces using MD simulations (Fig. 10). Their study focused on nanostructured surfaces with a combination of wet and dry regions to analyze how geometry and wettability affect condensation and heat transfer. The results demonstrated that hybrid (wet and dry) surfaces outperform homogeneous surfaces in terms of mass and heat transfer. The nanowet and nanodry pillars created Cassie droplets, which, due to their high mobility, significantly enhanced the condensation and heat transfer process. Additionally, it was observed that reducing the distance between the nanostructured pillars could limit the formation of Wenzel droplets, thereby increasing the condensation rate.

Paul and Hasan [63] used atomic simulations to investigate

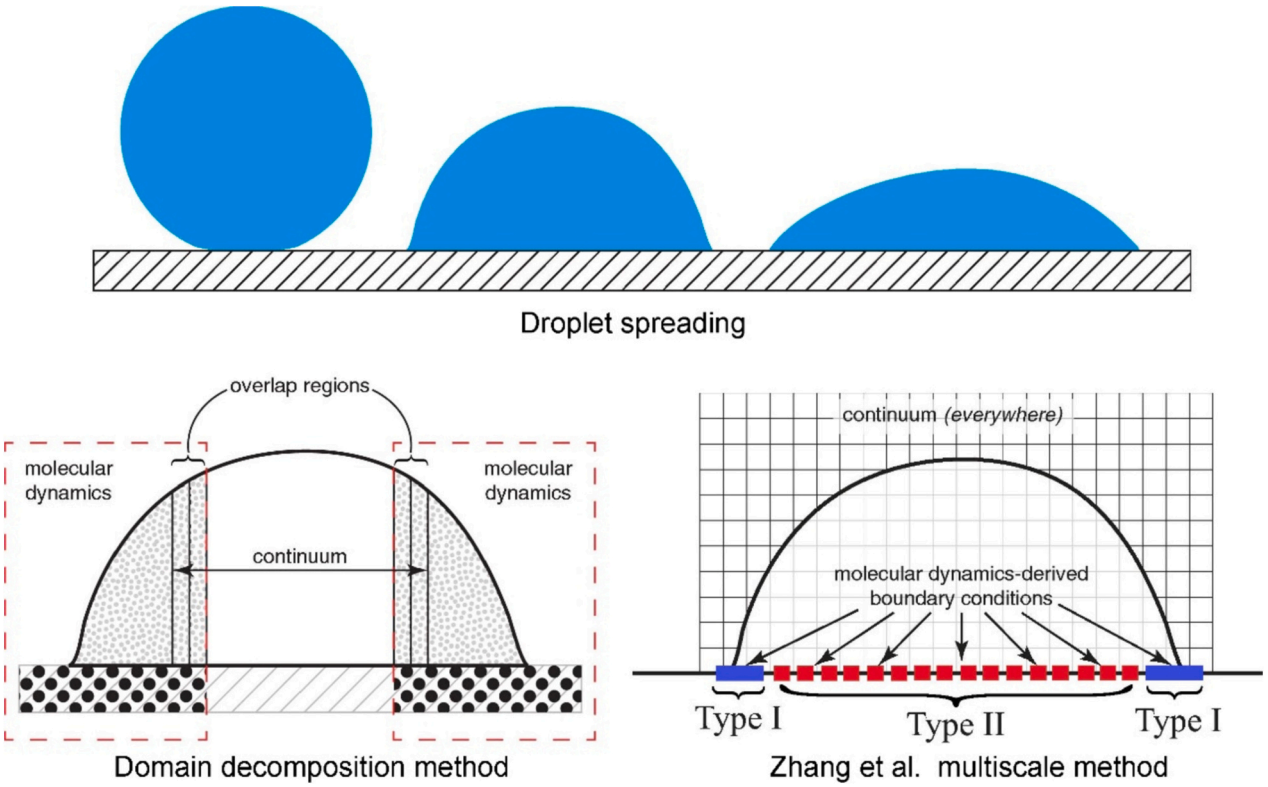

Fig. 8. Schematic diagrams analyzing the spreading of a nanoscale droplet on a surface and it various states [57].

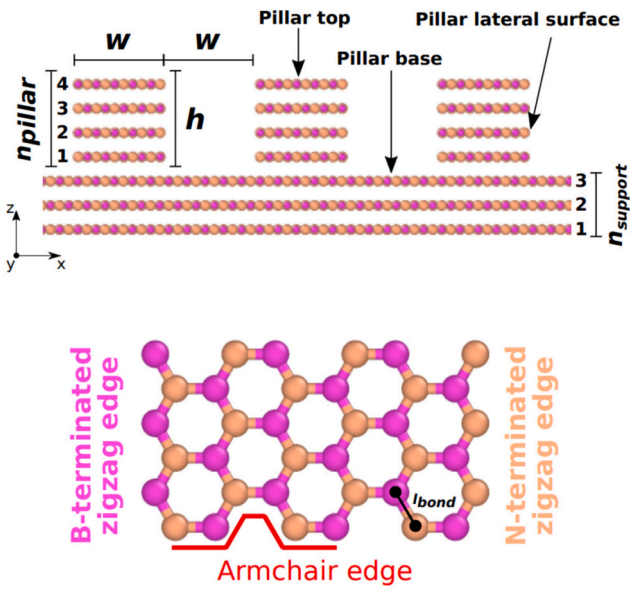

Fig. 9. Schematic representation of the surface of the nanostructure (Reproduced from ref. [60] with permission from The Royal Society of Chemistry).

condensation on hybrid surfaces with both gradient and patterned wettability. They found that gradient surfaces, which offer a smooth transition between hydrophilic and hydrophobic regions, outperform patterned surfaces in terms of droplet nucleation and condensation efficiency. This is because gradient surfaces promote continuous liquid movement, reducing the potential for droplet pinning, which is crucial for maintaining high heat transfer rates. Paul et al. [64] further demonstrated that combining strong hydrophilic sections with weak hydrophobic ones improves the condensation process, highlighting the importance of strategic wettability distribution in controlling nucleation and growth.

4.5. Wettability patterns and distribution

The distribution of hydrophilic and hydrophobic regions on hybrid surfaces can take different forms, either as patterned regions or gradient transitions. Patterned wettability refers to surfaces with distinct and well-defined regions of hydrophilic and hydrophobic areas, while gradient wettability involves a gradual transition between these two properties. Both approaches allow for controlled liquid movement and droplet behavior [56].

Paul et al. [64] showed that hybrid surfaces with a higher percentage of hydrophilic content significantly enhance condensation performance by promoting nucleation at hydrophilic regions. However, when the size of hydrophilic-hydrophobic strips is reduced, condensation occurs more rapidly due to the faster nucleation rate. Zhang et al. [65] provided insights into how surface roughness and low-energy coatings can create superhydrophobic properties, demonstrating that such surfaces have high contact angles and can repel water efficiently (Fig. 11). This principle of combining hydrophobicity with roughness can be extended to hybrid surfaces, where strategic combinations of hydrophilic and hydrophobic regions create surfaces with enhanced liquid control.

4.6. Hybrid wetting and energy states

The wetting behavior of hybrid surfaces is fundamentally driven by differences in surface energy between hydrophilic and hydrophobic regions. Hydrophilic regions tend to have lower surface energy, promoting droplet adhesion and spreading, while hydrophobic regions have higher surface energy, causing droplets to bead up and resist wetting. This difference in surface energy leads to the formation of distinct contact angles, which governs the movement and stability of droplets on the surface [58].

In hybrid wetting, the transition between the Wenzel and Cassie-

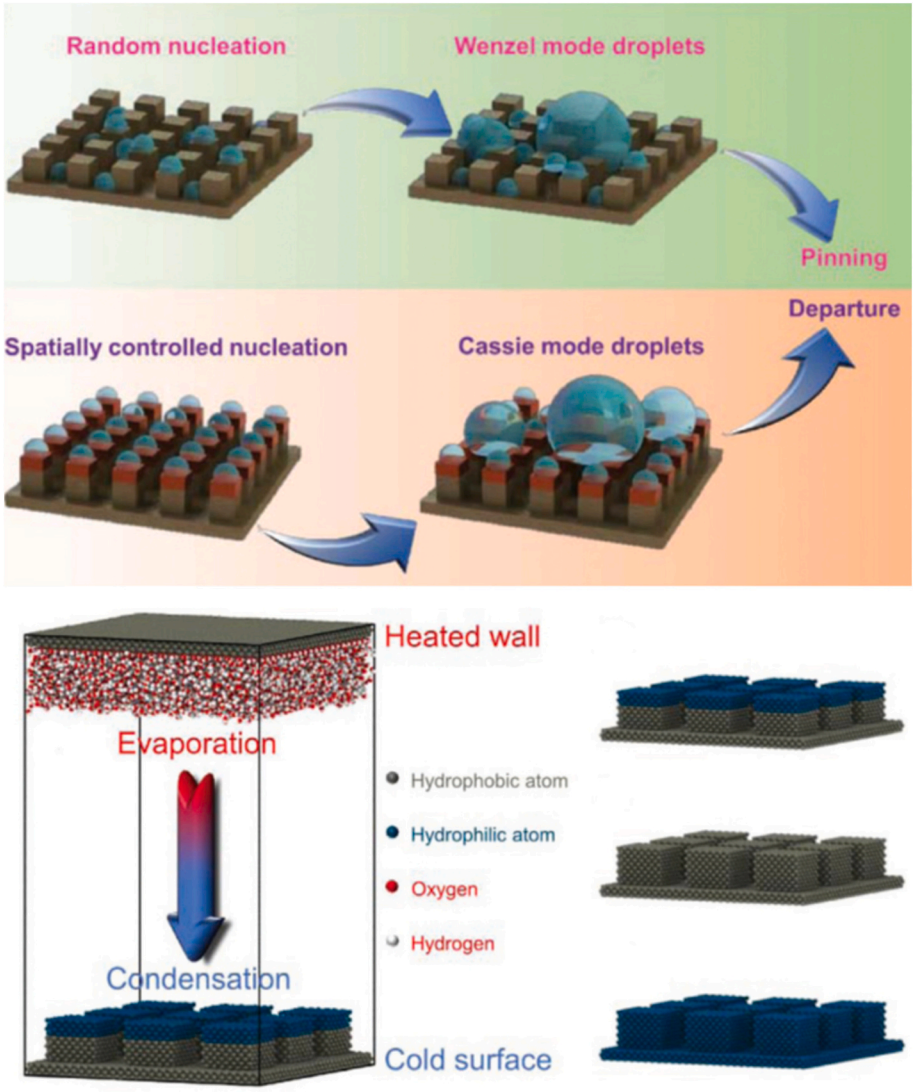

Fig. 10. Schematic of the condensation process on nanostructured surfaces presented by Gao et al. article (Reproduced from ref. [62] with permission from The Royal Society of Chemistry).

Baxter states plays a key role in determining droplet behavior. In the Wenzel state, surface roughness enhances wetting by increasing the surface area in contact with the liquid, leading to strong droplet adhesion. In contrast, the Cassie-Baxter state is characterized by air pockets trapped beneath the droplet, which reduces adhesion and promotes droplet mobility. Quéré [66] explored these transitions and their impact on surface wetting properties, showing that surfaces can switch between these states depending on the balance of surface tension forces. Patankar [67] examined superhydrophobic surfaces and the design of structures with dual roughness or tall pillars for developing self-cleaning surfaces, as illustrated in Fig. 12. Inspired by the lotus leaf's microscopic structure, known for its natural superhydrophobicity (the lotus effect), the article provided a theoretical and computational analysis of how to enhance water droplet contact angles on such surfaces. Patankar employed the Wenzel and Cassie-Baxter models to predict droplet contact angles on rough surfaces, demonstrating that using structures with dual roughness or tall pillars with a high length-to-width ratio could significantly increase the apparent contact angle. The dual roughness structure, as modeled in Fig. 12, shows how combining fine-scale roughness with coarse-scale elements further amplifies surface hydrophobicity, leading to lower droplet adhesion and improved water repellency.

4.7. *Dynamic wetting behavior and time dependence*

Hybrid wetting surfaces exhibit dynamic wetting behavior, where the interaction between liquid and surface changes over time. This is particularly important in processes such as condensation, where droplet formation, growth, and coalescence must be carefully controlled. Over time, the wetting properties of a surface can evolve, with droplets transitioning between wetting states or being influenced by factors such as surface roughness and temperature gradients [68].

Zhou et al. [69] investigated the wetting behavior and interfacial bonding of polymer-metal structures using MD simulations, as shown in Fig. 13. They showed how surface roughness and temperature affect the wetting dynamics of these materials. As temperature increases, the wettability of the polymer improves, allowing for better adhesion and interfacial bonding with the metal substrate. This dynamic interaction between surface structure and external conditions mirrors the behavior seen in hybrid wetting systems, where the balance of forces can shift over time, leading to changes in droplet behavior and surface interactions.

Rezaei et al. [70] also studied wetting behavior in membrane distillation systems and highlighted the importance of controlling surface wettability to prevent undesirable liquid infiltration. Although this

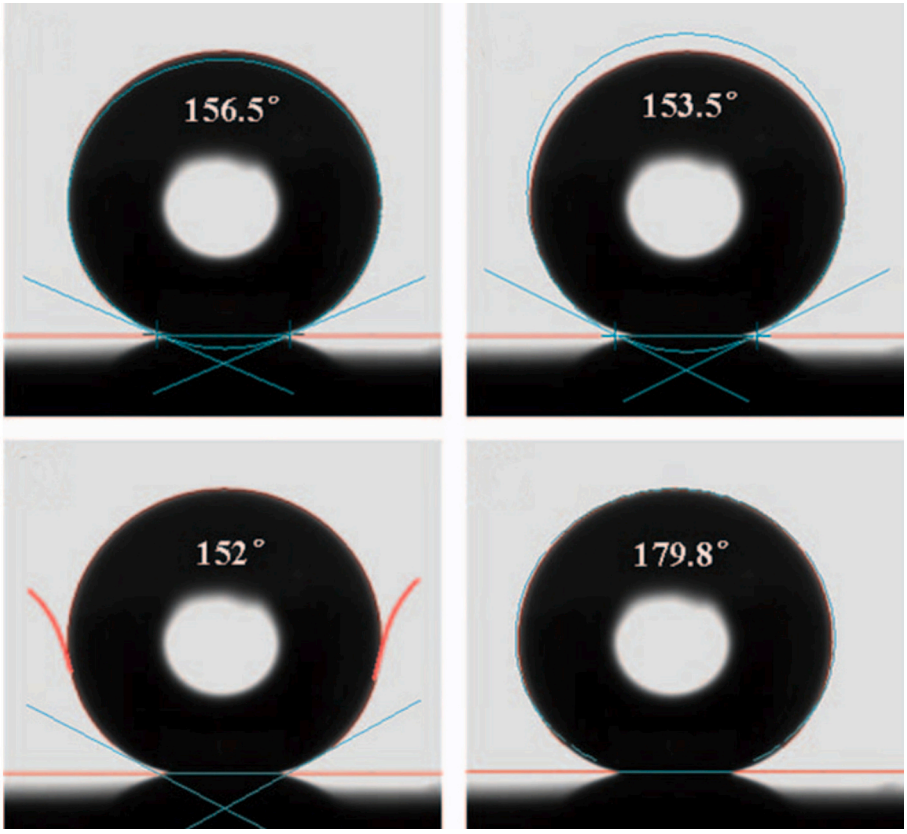

Fig. 11. Images of a water droplet on a superhydrophobic surface under different contact angle fitting conditions in various states (Reproduced from ref. [65] with permission from The Royal Society of Chemistry).

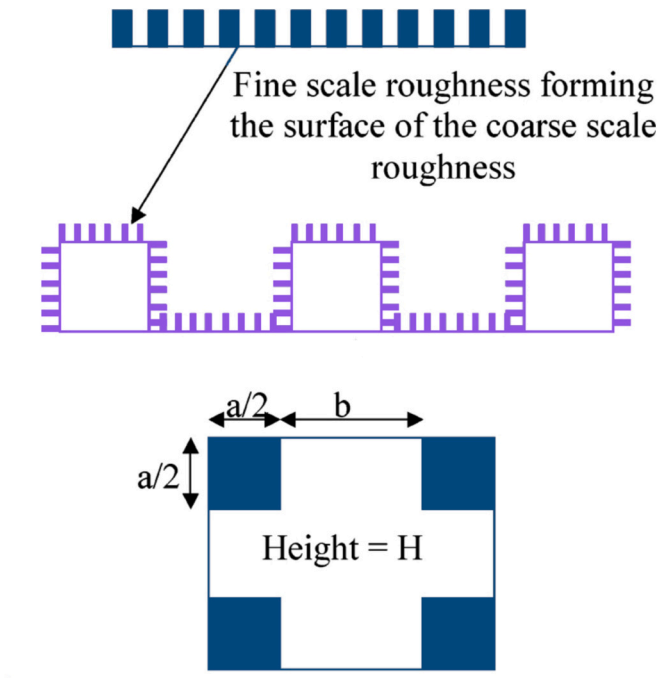

Fig. 12. Roughness geometry model for theoretical analysis in various states (Adapted with permission from [67]. Copyright 2004 American Chemical Society).

study focused on a specific industrial application, the principles of wetting behavior are broadly applicable to hybrid surfaces, where precise control over liquid movement is essential for maintaining system efficiency.

In summary, the research works discussed above have provided a comprehensive understanding of hybrid wetting, illustrating its significant impact on droplet behavior and surface interactions. To enhance clarity and aid in understanding the key findings, Table 1 presents a concise summary of these studies, highlighting the underlying mechanisms and concepts of hybrid wetting. This table serves as a useful reference, offering a comparative view of the contributions from different researchers and reinforcing the theoretical insights discussed in this section.

The findings summarized in Table 1 demonstrate that surface roughness and nanostructures have a substantial effect on the wetting properties of materials. Adjusting surface geometry, particularly by introducing dual roughness or nanocolumn structures, can greatly increase contact angles, leading to the creation of superhydrophobic surfaces with significant liquid repellency. These features are crucial for improving liquid behavior on hybrid surfaces. MD simulations have highlighted how increasing temperature and surface roughness can enhance wetting behavior and interfacial adhesion, especially in polymer-metal hybrid systems. These insights underline the importance of optimizing surface conditions, nanostructure designs, and temperature effects in various surface engineering applications.

Furthermore, these studies have explored the roles of nucleation and condensation on nanostructured surfaces, showing that surface morphology plays a critical role in improving condensation rates and heat transfer. For instance, reducing the distance between nanocolumns enhances condensation, providing new strategies for surface design in heat transfer applications. Multiscale simulation methods have emerged as powerful tools to study dynamic and microscopic phenomena on

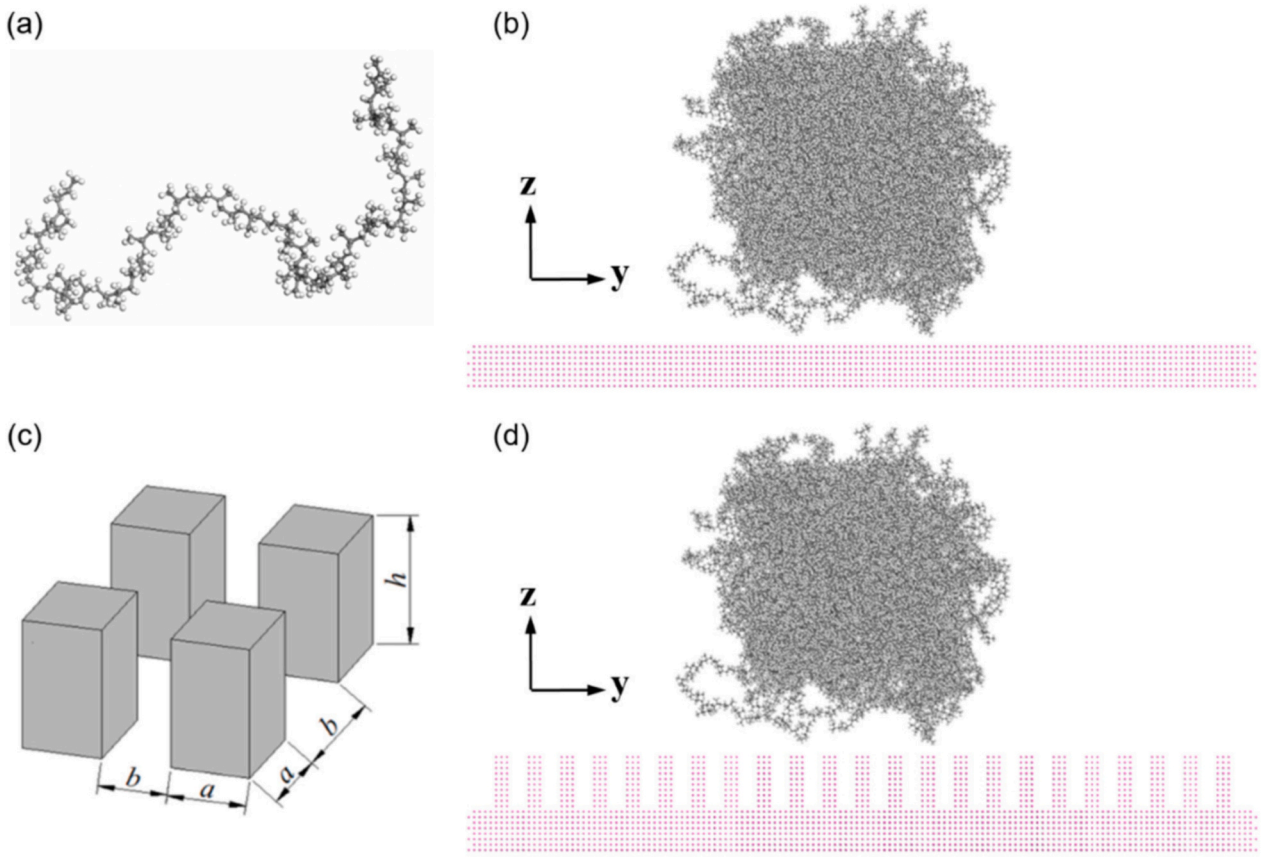

Fig. 13. Atomic models for the wetting behavior of PP droplets on a metal substrate [69].

hybrid wetting surfaces, though challenges remain in terms of achieving reproducibility and simplifying the transition of these models to industrial applications. Future research should focus on refining nanostructures to meet specific wetting requirements, improving simulation accuracy, and developing new methods for managing and recovering wet surfaces in membrane systems.

5. Hybrid wetting: applications and benefits

Hybrid wetting in nanochannels, as an emerging technology, has found wide applications in various scientific and industrial fields [72]. Due to their small size and high surface-to-volume ratio, nanochannels are greatly affected by surface properties such as wetting [73]. The combination of hydrophilic and hydrophobic regions on the internal surfaces of nanochannels can provide more precise control over fluid flow, improve heat transfer, and even optimize chemical processes. One of the main applications of hybrid wetting in nanochannels is improving heat transfer. Nanochannels are used in advanced cooling systems, especially in microelectronics. By using hybrid surfaces that feature both hydrophilic and hydrophobic areas, optimal conditions for droplet condensation can be created. This type of condensation reduces thermal resistance and increases heat transfer rates, improving the performance of cooling systems [74].

In addition to improving heat transfer, hybrid wetting can play an important role in optimizing fluid flow in nanochannels. In many applications, precise control over fluid flow in nanochannels is essential. Hybrid surfaces can direct fluids to specific areas and prevent liquid accumulation in undesirable regions, improving the efficiency and performance of microfluidic systems. This feature is particularly important in applications such as drug delivery in microfluidic devices [75,76]. In the field of separation and purification, hybrid wetting can serve as an effective tool for separating liquids with different properties. Nanochannels with hybrid surfaces can selectively attract or repel different liquids, improving separation and purification processes [77]. This application is particularly important in the oil and gas industry and in water purification. Another advantage of using hybrid wetting in nanochannels is reducing channel blockages. In systems that use nanochannels for fluid transfer, blockages can pose a serious problem. Hybrid surfaces can prevent the formation of liquid films and help maintain fluid flow along the nanochannel, improving the lifespan and efficiency of microfluidic systems [78]. Hybrid wetting also plays a role in improving chemical processes that take place in nanochannels. Due to their small size and large surface area, nanochannels provide an ideal environment for chemical reactions. Hybrid surfaces can more effectively control and optimize chemical reactions [52,79].

One of the emerging applications of hybrid wetting in nanochannels is the development of microfluidic devices for biological research [80]. In these devices, precise control over the movement and reaction of biological fluids such as blood and other biological samples is crucial. Hybrid surfaces can assist in better separation and analysis of samples, improving the accuracy of biological experiments [81]. This application is particularly important in the development of advanced diagnostic and therapeutic devices. The use of hybrid surfaces in nanochannels can also improve the efficiency of energy storage systems. In lithium-ion batteries and other energy storage systems, controlling electrolyte wetting in nanochannels can increase the capacity and lifespan of batteries [82]. This feature is particularly important in the development of the next generation of high-performance, long-lasting batteries. In the field of nanomedicine, nanochannels with hybrid surfaces can contribute to the development of targeted drug delivery systems. These systems can direct drugs to specific areas of the body and prevent drug accumulation in undesirable regions. Hybrid surfaces can improve this process by precisely controlling wetting, increasing the effectiveness of treatments [73,74]. Fig. 14 illustrates the applications of hybrid wetting, which will

Table 1
Summary of key research on the concepts and mechanisms of hybrid wetting.

Reference	Highlight	Remark	Benefits/ Limitations
Wagemann et al. [60]	Study of wetting behavior on nanostructured hexagonal boron nitride surfaces using MD.	Zigzag edges are more hydrophilic, while armchair edges are more hydrophobic. Altering edges modifies wetting.	Highlights tunable wetting via edge modification but lacks experimental validation for real-world applications.
Quere [61]	Examines the effect of surface roughness on wetting properties of materials using hybrid structures.	Surface roughness can enhance or prevent wetting properties, affecting system performance.	Provides insights into roughness effects but doesn't address scalability or durability in harsh environments.
Gao et al. [62]	Simulates water vapor condensation on hybrid nanostructured surfaces.	Hybrid surfaces enhance heat and mass transfer, with reduced nanocolumn spacing preventing Wenzel-mode droplets.	Demonstrates improved condensation but computationally intensive for large-scale simulations.
Zhang et al. [65]	Investigates superhydrophobic surfaces and their applications.	High contact angles and self-cleaning properties enhance industrial performance.	Effective but expensive to fabricate at industrial scale.
QUÉRÉ [66]	Analyzes Wenzel and Cassie-Baxter models for superhydrophobicity on nanostructured surfaces.	Higher contact angle hysteresis in Wenzel state increases adhesion and reduces self-cleaning.	Explores theoretical models but limited by practical constraints on transitioning states.
Patankar [67]	Investigates superhydrophobic surfaces with dual roughness inspired by lotus leaf structures.	Dual roughness increases contact angle and stabilizes composite state for better water repellency.	Effective design strategy but requires advanced fabrication techniques.
Zhou et al. [69]	Studies wetting behavior and interfacial bonding between polymers and metals using MD simulations.	Increasing temperature improves polymer-metal adhesion, mirroring hybrid surface dynamics.	Limited to specific polymer-metal combinations and temperatures.
Rezaei et al. [70]	Investigates wetting in membrane distillation processes.	Prevents liquid infiltration with hybrid wetting designs, improving membrane performance.	Addresses pore wetting issues but lacks long-term durability studies.
Zhang et al. [71]	Explores droplet spreading on surfaces using multiscale MD and VOF simulations.	Multiscale methods effectively model slip at the contact line and dynamic wetting phenomena.	Computationally demanding for large systems and extended time scales.

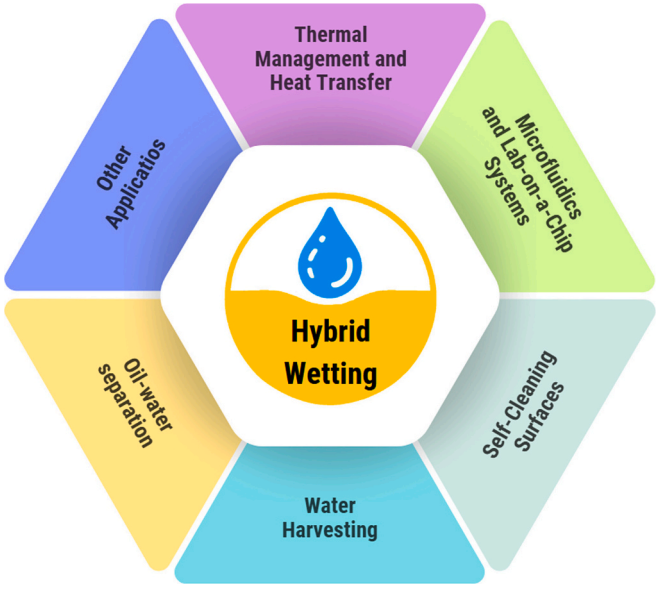

Fig. 14. Applications of hybrid wetting.

be followed by a review of previous studies related to each of them.

5.1. Thermal management and heat transfer

Hybrid wettability surfaces, combining hydrophilic and hydrophobic regions, have emerged as a key innovation in improving thermal management and heat transfer processes. These surfaces offer a distinct advantage by manipulating bubble formation, evaporation, and condensation at the liquid-vapor interface, which plays a crucial role in enhancing boiling heat transfer and evaporation efficiency. Studies using MD simulationshave demonstrated that hybrid surfaces can facilitate controlled bubble formation and increase heat flux, particularly on grooved surfaces with mixed wettability patterns. Other works have further shown that the variation in contact angles between hydrophobic patterns and substrates can significantly improve critical heat flux, optimizing boiling efficiency. The following literature review discusses the mechanisms uncovered in recent research, focusing on bubble dynamics, surface morphology, and wettability patterns as they relate to thermal management.

Zhao et al. [83] studied the mechanisms of bubble formation and heat transfer on grooved surfaces with hybrid stability (hydrophilic and

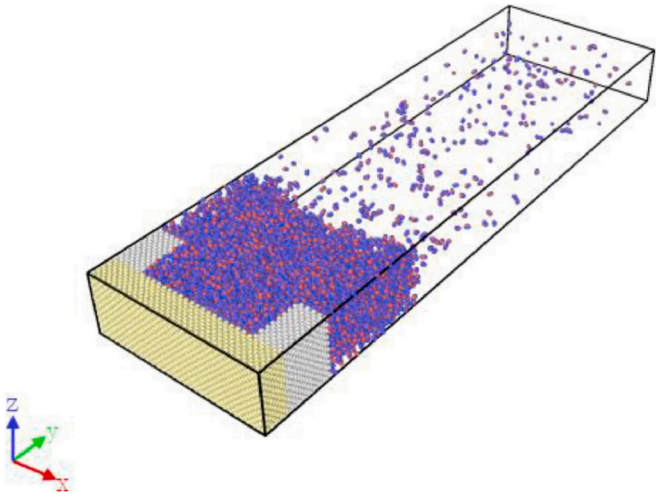

Fig. 15. The designed box in Zhao et al.'s work [83].

hydrophobic) under transient temperature boundary conditions (Fig. 15). In this research, MD simulations were conducted to describe the movement of the bubble contact line. The results showed that bubble growth on a fully hydrophilic surface was faster than on other grooved surfaces. Additionally, surfaces with a hybrid pattern, such as one with a hydrophilic bottom wall and hydrophobic side walls, could facilitate bubble formation while significantly improving the maximum heat flux. The innovation of this paper lies in examining the effects of hybrid stability on heat transfer and bubble formation. The findings of this research contribute to a deeper understanding of the mechanisms that enhance boiling heat transfer through bubble contact line movement on structured surfaces.

Ghani Dehkordi et al. [84] investigated the effect of an external electric field on the boiling behavior of H_2O/Fe_3O_4 nanofluid inside microchannels made of different materials using MD simulations. The simulations were performed using the LAMMPS software package, and key physical parameters such as kinetic energy, potential energy, radial distribution function, density, temperature, and the velocity profile of the nanofluid were analyzed. The results show that as the external electric field increases, the boiling process accelerates, with density increasing from 0.032 g/cm^3 to 0.038 g/cm^3, velocity up to 0.083 Å/ps, and temperature reaching 714 K. Phase transition (boiling) occurs earlier with the enhanced electric field, with the nanofluid boiling in platinum microchannels after 0.5 ns, in iron microchannels after 0.25 ns, and in gold microchannels after 0.21 ns.

Liang et al. [85] conducted experimental and modeling studies on heat transfer during pool boiling on surfaces with hybrid stability (hydrophobic and hydrophilic) (Fig. 16). These surfaces included hydrophobic dot and stripe patterns constructed on superhydrophilic substrates. The results show that these hybrid surfaces have a higher heat transfer coefficient compared to reference copper surfaces and hydrophilic or hydrophobic substrates. The innovation of the paper lies in modeling and examining the effects of contact angle variation between the patterns and the substrates on the critical heat flux. The study shows that increasing the contact angle difference between hydrophobic patterns and the substrate increases the critical heat flux but reduces the heat transfer coefficient in the nucleate boiling region. Additionally, the paper introduces a new theoretical model for predicting critical heat flux on hybrid stability surfaces, which has higher accuracy compared to existing models.

Cao et al. [86] investigated heat transfer due to liquid film evaporation on surfaces with hybrid stability at the nanoscale (Fig. 17). This research used MD simulations to study the effect of different ratios of hydrophilic and hydrophobic surfaces on heat transfer performance. The results show that increasing the hydrophilic surface ratio improves the heat transfer rate and evaporation rate. Additionally, thermal resistance at the solid-liquid interface decreases as the hydrophilic surface ratio increases, leading to significant improvements in heat transfer. On the other hand, the potential energy at the liquid-vapor interface shows minimal change with increasing hydrophilic surface ratio, but the liquid-vapor interface area increases, contributing to enhanced evaporation processes. This paper presents innovations in studying various combinations of hybrid surfaces and a more detailed examination of liquid-vapor interface behavior.

Peng et al. [87] investigated the effect of spherical surface roughness on the flow of argon atoms in microchannels under boiling conditions using MD simulations. With a constant wall temperature of 108 K and external forces ranging from 0.002 to 0.02 eV/Å, the results showed that spherical roughness has minimal impact on flow behavior, especially when the external force is between 0.002 and 0.01 eV/Å. Therefore, removing this type of roughness may not justify additional costs. In another study, Peng et al. [17] explored the effect of silver nanoparticles on the boiling behavior of argon flow in a cubic microchannel using MD simulations. With an external force of 0.002 eV/Å and a constant wall temperature of 108 K, the silver nanoparticles improved heat transfer and phase change, causing argon atoms to move from the walls to the center of the channel. Adding two, four, and six nanoparticles increased density by 20 %, 37 %, and 67 %, and temperature by 20 %, 49 %, and 84 %, respectively. While two nanoparticles had limited effect, four and six nanoparticles significantly improved heat transfer and are beneficial for practical applications.

Cai et al. [88] studied the effects of molar ratio and surface stability on the evaporation process of argon and krypton mixtures using MD simulations (Fig. 18). This research examined the influence of mixture composition and various surface stability conditions on phase transition. The results indicated that with an increased molar percentage of argon,

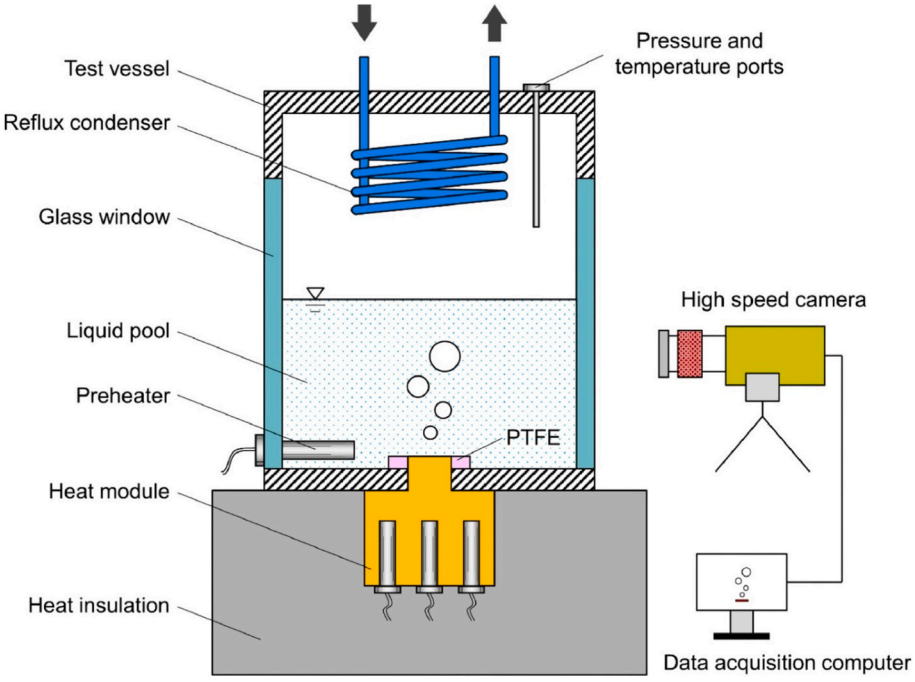

Fig. 16. The experimental setup of Liang et al. [85].

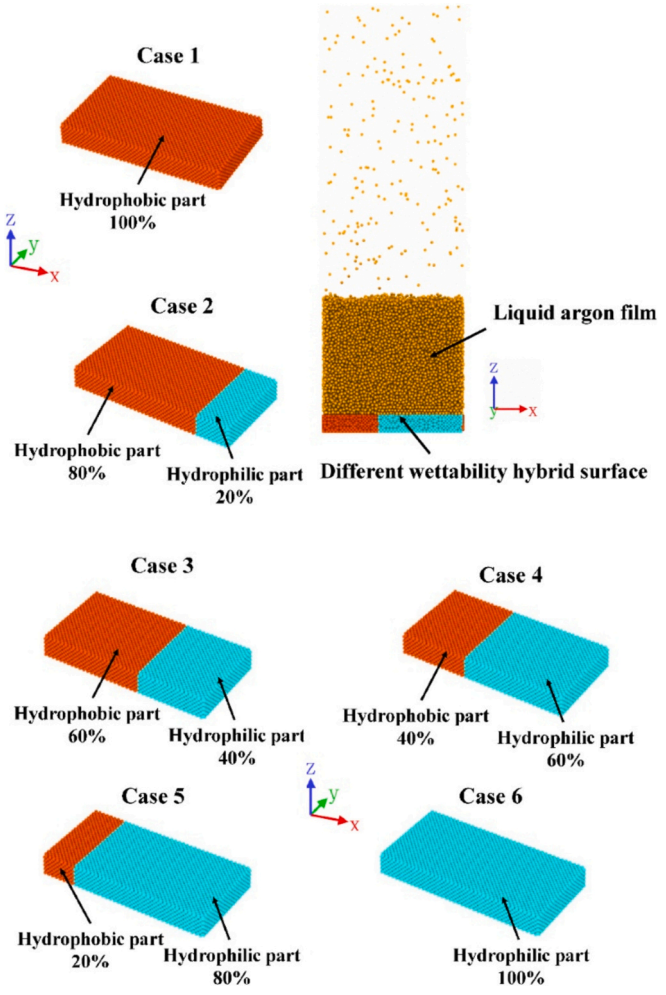

Fig. 17. Initial simulation system of liquid argon film on a hybrid wettability surface in the Cao et al.'s work [86].

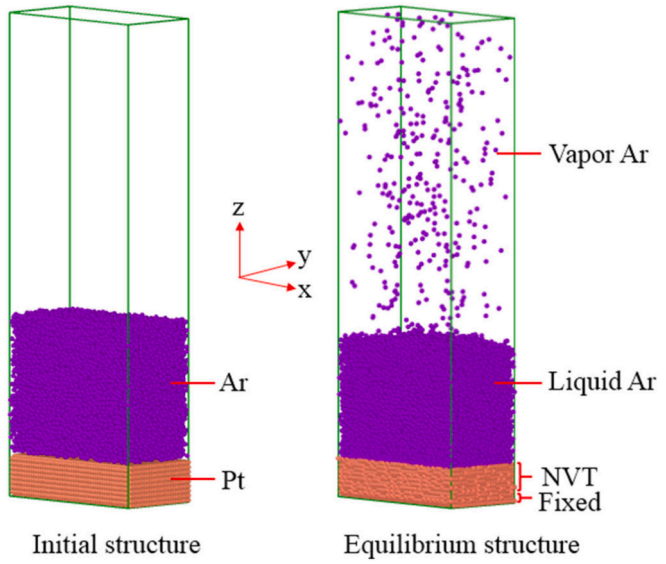

Fig. 18. Geometry studied in the simulation in Cai et al.'s study [88].

vapor film formation occurs, leading to reduced heat flux and decreased heat transfer. Adding krypton atoms to the system reduces or prevents vapor film boiling. It was also found that strongly hydrophobic surfaces result in faster bubble formation compared to weakly hydrophobic surfaces, and these initial bubbles enhance the evaporation process in systems with hybrid stability. In this study, the kinetic energy of fluid atoms and the interaction energy between platinum atoms and the fluid were used to explain the evaporation process, and the findings showed that a higher molar ratio of krypton decreases the likelihood of krypton atom evaporation.

Holder et al. [89] investigated the enhancement of heat transfer at low temperatures by using laser texturing on the inner surface of metal tubes. This study focused on improving dropwise condensation rather than filmwise condensation in the condenser region of two-phase heat transfer devices like closed-loop thermosyphons (Fig. 19). One of the main innovations of this research was the use of laser surface texturing and storing the tubes in a liquid containing long-chain hydrocarbons, which resulted in superhydrophobic surfaces with a contact angle of over 150 degrees. Stainless steel tubes, 1 m in length, were optimized using ultra-short laser pulses to create microscopic surface structures such as grooves and depressions. After storage and processing, these tubes were tested in the condenser section of thermosyphons. The results showed that the heat transfer rate for the laser-processed tubes increased to 0.92 kW at 45 °C, which was three times the heat transfer rate of unprocessed tubes (0.31 kW). The key finding of this study was that dropwise condensation, facilitated by hydrophobic surfaces, can significantly improve heat transfer performance, especially at low temperatures. At higher temperatures, this effect diminished, and heat transfer shifted to filmwise condensation. This research highlights the high potential of laser-processed tubes for industrial applications to enhance the efficiency of cooling and heat transfer systems.

Deng et al. [90] conducted a MD simulation study to investigate boiling heat transfer on thermally resistant surfaces with tunable wettability. The main innovation of this research lies in analyzing two states of surface wettability at different temperatures, involving hydrophilic surfaces at lower temperatures and hydrophobic surfaces at higher temperatures. The results show that these surfaces outperform those with fixed wettability, with an optimal range of wettability change that improves the boiling and heat transfer process. The study elucidates the mechanisms behind this process and offers suggestions for improving heat transfer systems.

Guo et al. [91] performed MD simulations to examine evaporation behavior on surfaces with varying wettability. The results indicate that surfaces with hybrid wettability enhance vapor film formation due to boiling heat transfer. Increasing hydrophilicity improves the vapor film production rate, while hydrophobicity boosts evaporation efficiency.

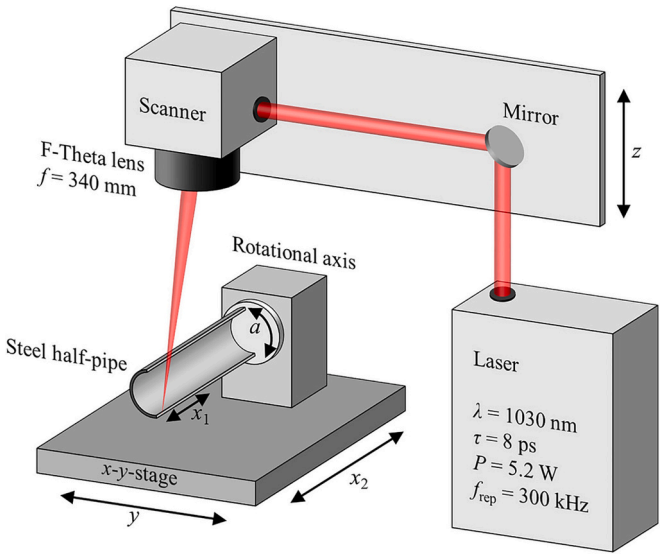

Fig. 19. Experimental setup in the study by Holder et al. [89].

The study also explored optimal designs for hybrid wettability surfaces, demonstrating that at an optimal ratio of 0.64 and four wettability structures, vapor film production increased by 60 % compared to fully hydrophobic surfaces, and evaporation efficiency improved by 33.8 % compared to fully hydrophilic surfaces.

Hossain et al. [68] investigated liquid-to-vapor phase transition on nanoporous surfaces with different wettability conditions and surface morphologies. Using non-equilibrium MD simulations, they evaluated the heat transfer and evaporation characteristics in a system containing liquid and vapor argon on a platinum nanoporous substrate. The results showed that nanoporous structures significantly improved heat transfer compared to flat surfaces. Bubble nucleation sites and growth patterns depended on pore height and wettability conditions. Additionally, as pore height increased, heat accumulation also increased, and surfaces with hydrophilic bases and hydrophobic substrates performed similarly to fully hydrophilic surfaces.

Zhang and Xu [92] used MD simulations to investigate nucleate boiling on nanostructured surfaces with hybrid wettability. The results revealed that hydrophobic nanopores in solid walls enhance the boiling process. Residual vapor nuclei in these nanopores create a piston effect at the vapor-liquid interface, increasing internal vapor pressure. Additionally, contact angles less than 18 degrees in the nanopores cause bubbles to form only within these nanopores, whereas bubbles tend to form on flat surfaces for hydrophilic surfaces. These findings contribute to optimizing the design and performance of micro/nano systems.

Li et al. [93] explored surface wettability control and fluid flow guidance in various natural and engineering contexts. This review categorizes recent advances, focusing on the structures and operating principles of surfaces with directional wettability. It also examines different strategies researchers have used to manipulate fluid wettability behavior through passive and active control mechanisms. Surfaces with passive wettability do not require additional energy and allow for continuous, broad fluid control, whereas surfaces with active wettability provide more precise control and greater flexibility.

In summary, hybrid wettability surfaces represent a promising innovation in enhancing thermal management and heat transfer, particularly in processes involving boiling and evaporation. The ability to manipulate bubble dynamics, improve heat flux, and optimize phase change processes through tailored surface patterns has far-reaching implications for a range of applications. From MD simulations to experimental validations, these surfaces have demonstrated significant improvements in heat transfer efficiency. As research in this area continues to evolve, hybrid wettability surfaces are poised to play a critical role in developing more efficient thermal systems, including next-generation heat exchangers, cooling technologies, and energy-efficient devices.

Hybrid wettability surfaces have demonstrated significant improvements in thermal management applications. For instance, surfaces combining hydrophilic and hydrophobic regions show up to 10× higher heat transfer efficiency in dropwise condensation compared to filmwise condensation on purely hydrophilic surfaces. Furthermore, optimized ratios of hydrophilic to hydrophobic regions can reduce thermal resistance by 20–30 %, as shown in MD simulations. Liang et al. [85] reported that hybrid patterned surfaces with increased contact angle variation achieved up to 15 % higher critical heat flux compared to conventional hydrophilic surfaces. Similarly, surfaces with nanostructures have improved nucleate boiling heat transfer rates by up to 40 %, enabling more efficient cooling in electronic systems. However, challenges persist, particularly in scaling these designs to large systems, where maintaining consistent wettability properties and ensuring robust surface fabrication remain significant hurdles. For industrial applications, further research is needed to translate these nanoscale improvements into macro-scale heat exchanger designs.

5.2. Microfluidics and lab-on-a-chip systems

Microfluidics and lab-on-a-chip systems are revolutionizing the control of fluid behavior at the micro- and nanoscale, offering numerous applications in areas such as biomedical diagnostics and environmental monitoring. These systems often require precise manipulation of liquids, where the interplay of surface properties and droplet behavior becomes critical. Recent research highlights how the geometry of droplets, surface structures, and temperature gradients play key roles in fluid dynamics within nanochannels. Advances in hybrid wetting surfaces, which combine hydrophilic and hydrophobic characteristics, have shown promise in optimizing evaporation, condensation, and fluid stability. This section explores these innovations, focusing on how hybrid wetting can enhance performance in microfluidic applications.

Hiratsuka et al. [94] studied the effects of droplet geometry and size on evaporation behavior at the nanoscale. They used MD simulations to study water droplets measuring 5 nm and 6 nm, consisting of 2259 and 3787 water molecules, respectively. One of the innovations of this research is the investigation of small droplet size effects on evaporation and bubble formation in nanochannels (Fig. 20). The results show that larger droplets take longer to evaporate, while bubble formation occurs more quickly in smaller droplets. Additionally, the surface geometry on which the droplets rest has a significant impact on evaporation and droplet stability. This paper demonstrates that small changes in surface geometry can substantially influence the heat transfer and evaporation process.

Xu and Chen [95] investigated the process of water vapor condensation on wedge-shaped surfaces with varying wettability gradients (Fig. 21). In this study, MD simulations were used to examine the effect of apex angles and different wettability gradients on the drainage rate of condensates. The innovation of this research lies in the study of surfaces with multiple wettability gradients, categorized as a combination of hydrophilic (water-attracting) and hydrophobic (water-repelling). Three groups of surfaces were examined: hydrophilic to superhydrophilic, hydrophobic to hydrophilic, and hydrophobic to superhydrophobic. The results showed that filmwise condensation occurred in the first group (hydrophilic to superhydrophilic), but the drainage rate was very low. In the second group (hydrophobic to hydrophilic), droplet condensation occurred, where droplets could be controlled by the surface and moved downward, improving drainage efficiency. In the third group (hydrophobic to superhydrophobic), no condensation was observed. The results also indicated that in the second group, a 14-degree apex angle provided the best performance for accelerating condensation and condensate drainage. Overall, this study demonstrates that wedge-shaped surfaces with varying wettability gradients can significantly improve condensate drainage at the microscale.

Li et al. [96] investigated the fusion process of liquid nanoparticle droplets at different temperatures using MD simulations (Fig. 22). The innovation of this research lies in examining the effect of temperature on the fusion process and the atomic diffusion between two metallic (Pb) nanoparticles. The results showed that increasing temperature

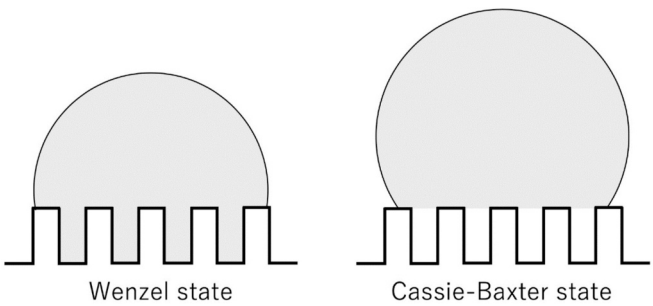

Fig. 20. Schematic diagrams of the Wenzel and Cassie-Baxter states in Hiratsuka et al.'s study [94].

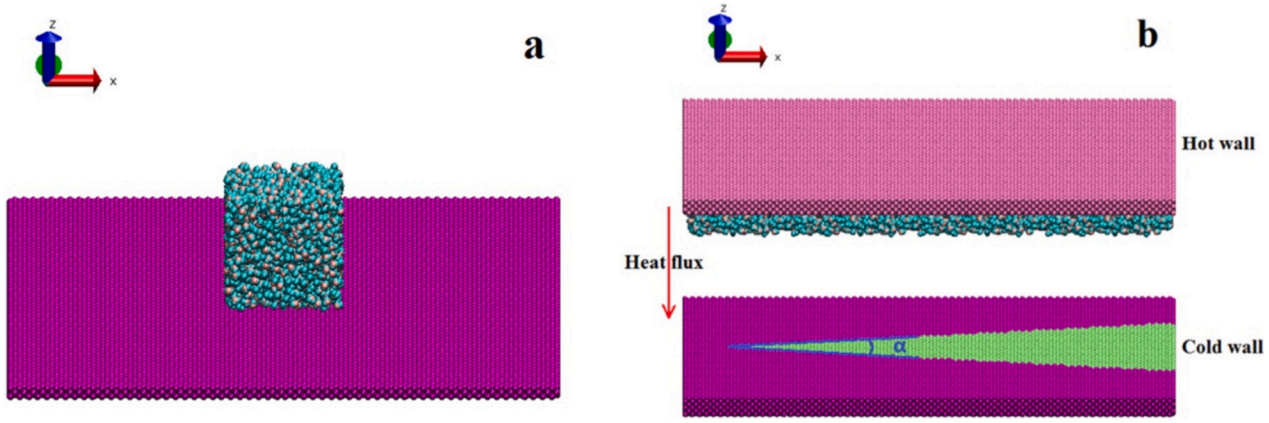

Fig. 21. (a) Initial water model on the copper surface. (b) Water condensation model on a wedge-shaped composite surface in Xu and Chen's study [95].

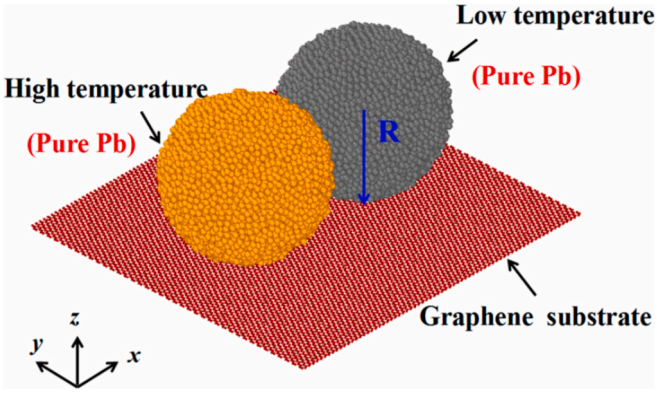

Fig. 22. Schematic of the fusion system for two nanodroplets at different temperatures in Li et al.'s study (Reprinted with permission from [96]. Copyright 2019 American Chemical Society).

significantly improved the droplet fusion rate, due to the increased atomic diffusion and improved mixing process. This study identified three different models of nanoparticle fusion: uniform model, separation model, and coating model, each depending on the temperature difference between the two droplets. The findings indicated that the greater the temperature difference between the nanoparticles, the faster and more efficient the fusion process. The research also suggested that selecting a metal with a higher temperature tolerance could further improve the fusion process.

Sun et al. [97] examined the effects of surface wettability and topology on liquid flow in micro and nanochannels using a hybrid MD-continuum simulation method (Fig. 23). This multiscale study investigated the effects of channel height, the shape of rough surface elements, the ratio of step-to-element length, and liquid-solid bond strength on the velocity and temperature boundary conditions. The results showed that velocity boundary conditions changed from significant slip to lock-in due to obstruction by the nanoscale surface structures. Distorted flows and density oscillations near the wall were also observed for small step ratios and strong liquid-solid bonding. The study concluded that the relative slip length and Kapitza length changed approximately linearly with liquid-solid bonding strength and channel height. An important finding was that the shape of surface roughness elements had less influence compared to other parameters.

Kwon et al. [98] introduced an innovative microfluidic system for detecting and counting nanoparticles in the air, using water as the condensing liquid. This system was designed to accurately assess exposure to airborne nanoparticles. In this study, a microfluidic counter based on the water vapor condensation mechanism was developed,

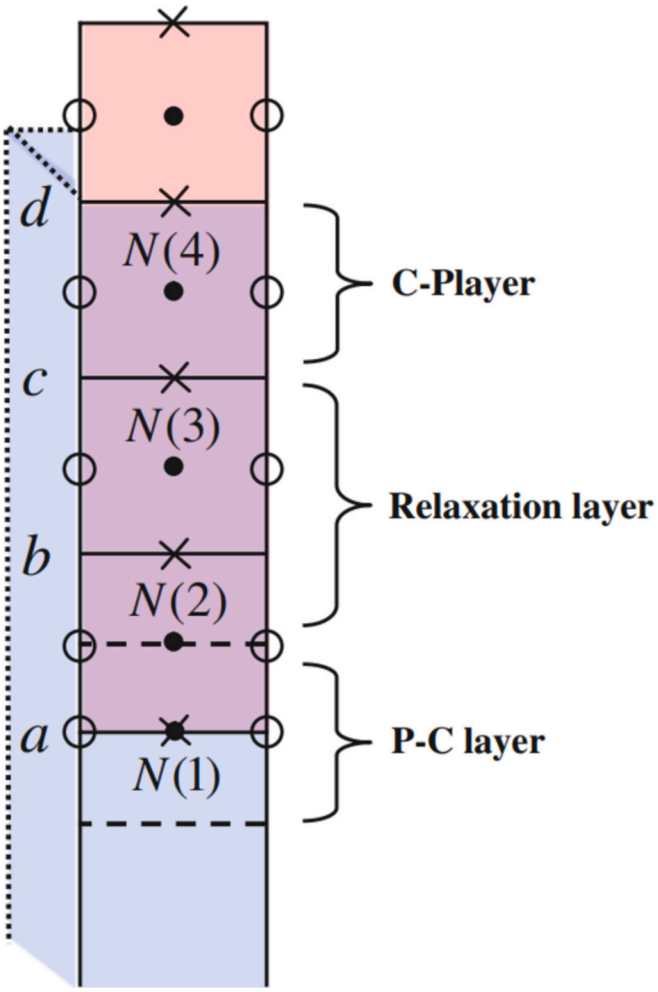

Fig. 23. Schematic of the overlap region configuration in Sun et al.'s study [97].

featuring a specific pathway and C-shaped electrodes that enabled precise measurement of nanoparticles suspended in the ambient air. This device, utilizing a 3D-printed chip and cooling module, could detect nanoparticles ranging from 7 to 100 nm in size, with concentrations between 10^6 and 10^7 particles per cubic centimeter (Fig. 24). Using an electrostatic aerosol meter to compare the system's performance with existing commercial models, the results showed that the proposed microfluidic system could accurately count nanoparticles, sometimes

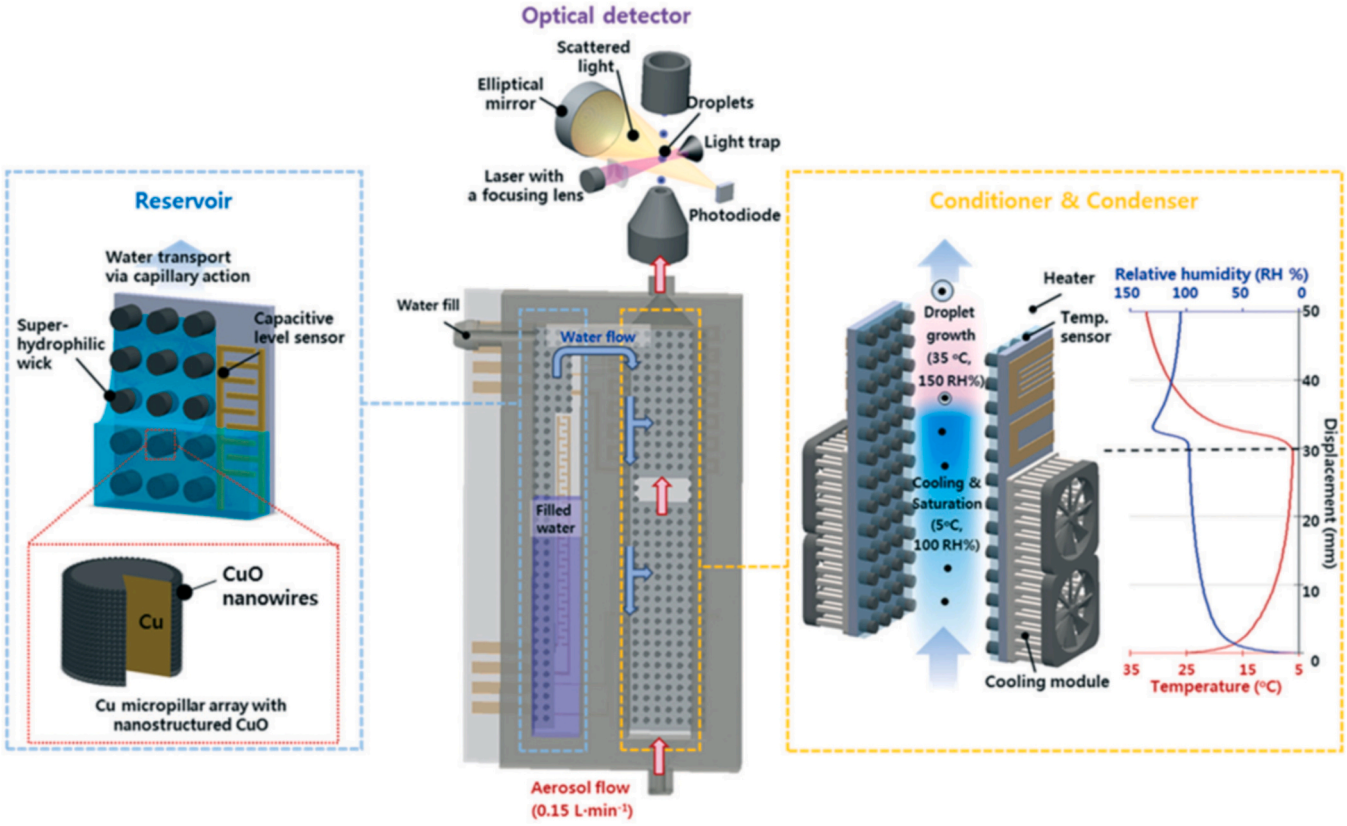

Fig. 24. Laboratory setup in Kwon et al.'s study [98].

exceeding 100 % counting efficiency due to fluctuations in reference concentration.

Chen et al. [99] proposed a simple and rapid method for creating superhydrophobic surfaces with self-cleaning properties. By immersing copper substrates in a solution containing zinc oxide, stearic acid, and polyphenylene sulfide, surfaces with a water contact angle of 164.7 degrees and a sliding angle of 4 degrees were created. MD simulations revealed that hydrophobic interactions between alkyl chains and hydrophilic interactions between water droplets and the zinc oxide surface were responsible for creating these properties. The study demonstrated that the combination of surface chemistry and surface roughness leads to superhydrophobic properties and excellent self-cleaning performance.

Qiang et al. [100] conducted MD simulations to investigate the mechanism of vapor condensation and suspended droplet formation on hierarchically nanostructured surfaces. The results showed that hierarchical structures facilitated the rapid formation and high mobility of suspended droplets during the condensation process. The study explored the effect of solid fraction and secondary structure distribution on the stability and mobility of suspended droplets, showing that increasing these parameters improved the stability of suspended droplets. Secondary structures on surface walls played an important role in creating suspended droplets, and these findings could help optimize the design of anti-icing and self-cleaning surfaces.

Ren et al. [101], using MD simulations, studied the spontaneous jumping of nanodroplets on hydrophobic surfaces with nanocone arrays. This nanostructured design caused nanodroplets to spontaneously jump upward after coalescence. Reducing the apex angle of the nanocones decreased the critical contact angle, with droplet jumping occurring at around 100 degrees. The study also showed that the initial position of the nanodroplets significantly affected the jumping speed; when nanodroplets were positioned between the nanocones, the jumping speed increased by up to 245 %. This study confirmed the importance of nanostructures in enhancing the efficiency of spontaneous droplet jumping and provided strategies for improving controlled droplet movement on functional surfaces.

Hao et al. [102] developed a robust superhydrophobic coating using multi-walled carbon nanotubes modified by octyltrimethoxysilane through a simple one-step spray method. The results showed that this coating not only had excellent self-cleaning properties but also exhibited resistance to acidic and alkaline corrosion. MD simulations examined the mechanism of coating modification and wettability, revealing that hydroxyl groups formed hydrogen bonds with the surface of the carbon nanotubes, while molecular tails extended away from the surface. After modification, the nanotube surface changed from hydrophilic to hydrophobic. Even after 400 cm of abrasion, the coating remained superhydrophobic.

The incorporation of hybrid wetting principles into microfluidics has led to significant improvements in managing fluid behavior at both the nano- and microscale. By manipulating surface properties, such as wettability gradients and nanostructured designs, these systems achieve enhanced evaporation control, more efficient droplet condensation, and better nanoparticle detection. Furthermore, the ability to fine-tune surface topologies has opened up new possibilities for optimizing liquid flow and droplet stability, which are critical for the functionality of lab-on-a-chip devices. These advancements demonstrate the potential of hybrid wetting to transform microfluidic systems, driving further innovation in applications ranging from diagnostics to material processing.

In microfluidics, hybrid wettability surfaces have proven highly effective for fluid manipulation, particularly in lab-on-a-chip systems. Xu and Chen [95] demonstrated that wedge-shaped surfaces with hydrophobic-to-hydrophilic gradients significantly improved condensate drainage efficiency, reducing flow resistance by up to 50 %. However, scaling these systems for high-throughput applications is limited by manufacturing constraints. Nanostructured surfaces require precise fabrication techniques, such as nanosphere lithography or chemical

etching, which are cost-intensive and difficult to standardize for large-scale production. Additionally, the high surface-to-volume ratio in microchannels amplifies issues such as clogging and liquid accumulation, particularly when working with biological samples or complex fluids. Ensuring the uniformity of wettability patterns across a large number of devices is another critical challenge that limits reproducibility in commercial applications. Advanced 3D printing techniques and modular designs may offer scalable solutions, but their feasibility for widespread use remains under investigation.

5.3. Self-cleaning surfaces

Self-cleaning surfaces have attracted significant attention in recent years due to their potential applications in various industries, including textiles, automotive, and electronics. These surfaces are engineered to reduce the need for cleaning by allowing water droplets to easily roll off, carrying away contaminants. The design of these surfaces is often inspired by nature, where features like lotus leaves or insect wings exhibit superhydrophobicity, enabling dirt removal without external assistance. The underlying mechanisms behind self-cleaning properties are closely linked to surface wettability and the interaction of droplets with surface micro- and nanostructures. This section reviews key studies that explore how hybrid wetting principles can be applied to enhance self-cleaning performance in various scenarios.

Zhang et al. [103] studied the self-cleaning mechanism of rough hydrophobic surfaces and showed that wettability transitions caused by droplet coalescence play a key role in this process. In this research, mass density particle dynamics simulations were used to study the wettability behavior of droplets on mechanically textured surfaces (Fig. 25). The simulation results showed that droplets in the Cassie-Baxter state had lower adhesion and hydrodynamic resistance than in the Wenzel state. The innovative aspect of the paper was the observation of a wettability transition from the Wenzel to the Cassie-Baxter state after droplet coalescence, driven by the energy released during coalescence. These findings contribute to the design of biomimetic self-cleaning surfaces and suggest methods for controlling liquid behavior at small scales through multi-droplet coalescence. Furthermore, the study showed that adding another droplet to the coalescence process could increase droplet jumping speed by an order of magnitude.

Pu et al. [104] investigated the growth and spontaneous jumping of condensed droplets on hydrophobic nanostructured surfaces. This study utilized MD simulations to explore the dynamics of droplet growth and jumping (Fig. 26). The innovation of this study lies in the analysis of Laplace pressure, which causes droplets to jump due to the pressure difference between the top and bottom of the droplet. The results showed that local hydrophilic pinning surfaces play a significant role in the growth and jumping process. These surfaces initially facilitate

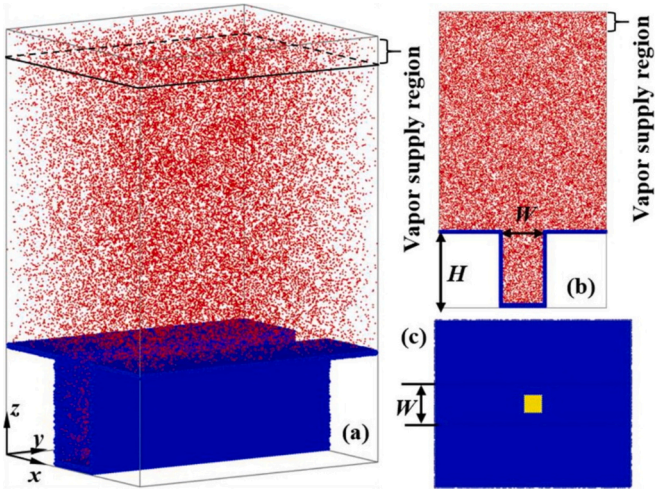

Fig. 26. Schematic of the problem investigated by Pu et al. [104].

droplet growth, but as pinning force increases, droplet jumping is restricted. In the first stage, the droplet grows on the hydrophobic surface and deforms. Then, as curvature difference and Laplace pressure increase, a force is generated that causes the droplet to detach from the hydrophilic pinning surface. Droplet jumping occurs when this force exceeds the pinning force. The study demonstrates that increasing the wettability or size of the pinning area accelerates droplet growth but makes jumping more difficult.

Ding et al. [105] explored the transition of water from a wet to a non-wet state on nanostructured surfaces with various wettability patterns. This study, using MD simulations, examined parameters such as volume, slope angle, and hybrid wettability ratios to study condensation and heat transfer processes (Fig. 27). The results showed that increasing the volume ratio improves water transport and condensation performance. Nanocone surfaces perform better in droplet shedding than nanopillar surfaces, and a slope angle of 10 degrees provided the best performance for facilitating droplet shedding. Additionally, a hybrid wettability ratio of 1/3 was identified as the optimal ratio for preventing droplet adhesion to surface textures. The key innovation of this paper lies in the optimal combination of these three parameters, improving surface performance without reducing condensation efficiency. The findings can help design superhydrophobic surfaces for applications like thermal management, water collection, and desalination.

Wisdom et al. [106] studied the self-cleaning mechanism of superhydrophobic surfaces through spontaneous motion of condensation droplets. This research demonstrated that water droplets formed from

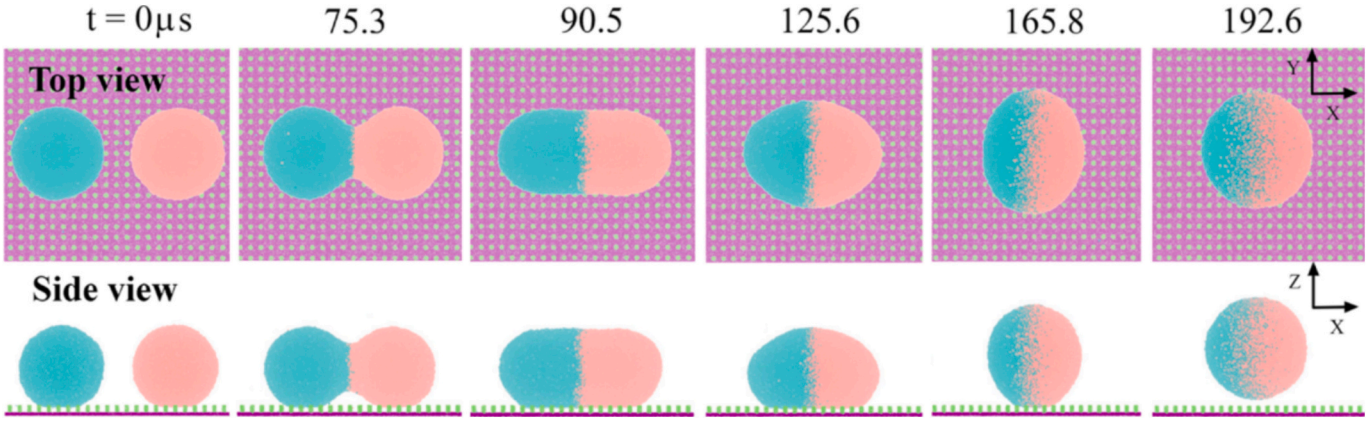

Fig. 25. Snapshots of the dynamics of two droplet coalescence in the Cassie-Baxter state, showing the jumping motion of the merged droplet from the textured surface in Zhang et al.'s study (Reprinted with permission from [103]. Copyright 2019 American Chemical Society).

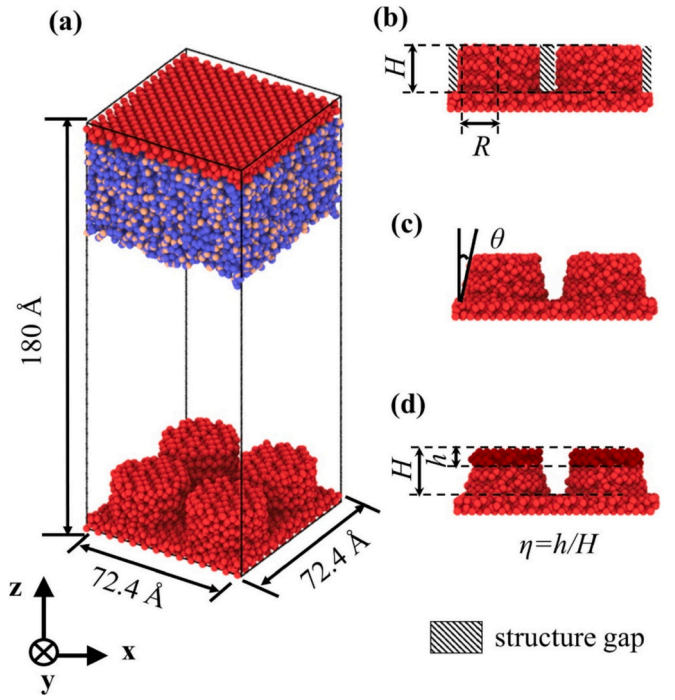

Fig. 27. Schematic diagrams for (a) the simulation domain, (b) parameters used in the examination of volume fraction, (c) parameters used in the investigation of tilt angle, and (d) parameters used in the examination of hybrid wettability ratio used by Ding et al. [105].

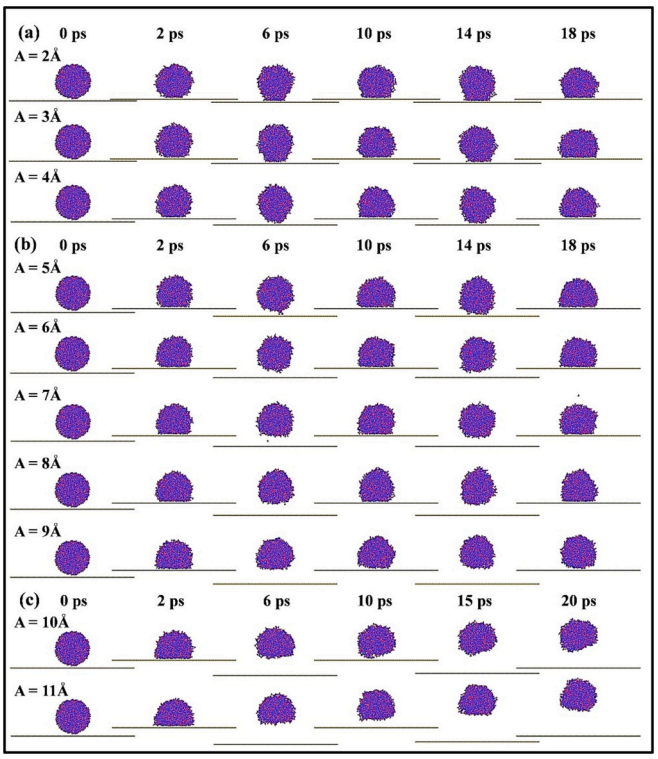

Fig. 28. Schematics of wetting systems under different amplitudes when the vibration amplitude of the graphene substrate is 8 ps in the work presented by Wang et al. [107].

vapor condensation on superhydrophobic surfaces can self-detach and carry away contaminants without external forces like gravity or wind. The innovation of this study is the discovery of the spontaneous droplet jumping mechanism, which helps clean surfaces such as insect wings and synthetic materials. A study on cicada wings revealed that their nanostructures and wax coatings create high contact angles and facilitate droplet condensation. The condensed droplets quickly coalesce and, due to surface energy released during coalescence, jump off the surface. This novel mechanism, unlike traditional self-cleaning methods that require rain or external forces, can operate in moist conditions without rain.

Wang et al. [107] examined the effect of vibrational fields on the wettability of graphene surfaces using MD simulations. The study specifically analyzed how sinusoidal vibrations influence the contact angle of water droplets and the transition between wetting states (Fig. 28). The primary innovation of the paper is its investigation of the controlled dynamics of surface wettability under vibration, which could help regulate surface hydrophobicity. The results indicate that increasing vibration amplitude raises the droplet contact angle, enhancing the surface's hydrophobicity. Three wetting states were identified: full wetting, transition between wetting and non-wetting, and complete non-wetting, which change depending on vibration amplitude. One key finding is that at low amplitudes, droplets stick to the surface, but as the amplitude increases, droplets begin to detach. Vibrations can also alter droplet adhesion and sliding behavior, causing transitions from the Wenzel to the Cassie state.

Miller [108] examined the self-cleaning characteristics of superhydrophobic surfaces through rolling and condensation of water droplets. The key innovation of the paper lies in comparing two methods—droplet rolling and condensation—for cleaning surfaces contaminated with salt, tobacco, and pollen. The results showed that salt-contaminated surfaces could be completely cleaned with several water droplets, restoring the contact and sliding angles to their original state. In the case of tobacco-contaminated surfaces, complete cleaning was not possible, and only partial contaminants were removed. Additionally, pollen contamination, due to hydrodynamic osmosis and cell rupture, resulted in the loss of superhydrophobic properties. The paper concludes that both droplet rolling and condensation methods can effectively clean superhydrophobic surfaces, but their efficiency depends on the type of contamination.

Dong et al. [109] investigated the interaction of nanoparticles with controlled wettability surfaces using MD simulations. Unlike macroscopic fluid transport, which occurs on hydrophilic or superhydrophobic surfaces, at the nanoscale, this transport is only possible on hydrophobic or superhydrophobic surfaces. The difference lies in the transport mechanism: at the nanoscale, viscosity forces are enhanced, and transport occurs through droplet jumping rather than ejection. The normalized transport velocity (V), which depends on contact angle differences and impact velocity, was extracted from simulation results, showing that optimal fluid transport at the nanoscale includes high impact velocity with large contact angle differences and low velocity with moderate contact angle differences. Finally, a simple energy conversion model was proposed, describing the relationship between V and $\Delta\theta$.

Song et al. [110] have studied the development of surfaces with controllable wettability. Such surfaces have broad applications in areas like corrosion resistance, anti-fogging, anti-icing, self-cleaning, reducing fluid friction, oil-water separation, and water harvesting. The study analyzed various methods to alter the chemical composition and microstructures of material surfaces using external stimuli such as light, temperature, pH, and electric potential. Additionally, processes like ion-beam etching, photolithography, particle hybridization, and nanopattern growth were investigated for fabricating surfaces with different wettability characteristics. Finally, the paper discusses the challenges and issues that need to be addressed in the future to improve these technologies.

In conclusion, self-cleaning surfaces represent a promising avenue for creating low-maintenance, efficient designs for various industries. Advances in understanding droplet behavior, such as coalescence-induced jumping or the influence of surface structures, have helped

drive innovation in this field. By manipulating surface wettability at micro- and nano-scales, engineers can create surfaces that not only repel water but also actively remove contaminants. As research continues, the development of hybrid wetting technologies will play a vital role in overcoming existing challenges, such as improving performance in varied environmental conditions and extending the longevity of these surfaces.

5.4. Water harvesting

Water scarcity is a critical challenge faced by many regions around the world, particularly in arid and semi-arid areas. As a result, water harvesting technologies that capture water from the atmosphere have garnered significant interest due to their potential to provide sustainable water sources. Hybrid wetting surfaces, which combine both hydrophilic and hydrophobic regions, play a crucial role in enhancing the efficiency of water collection from air, fog, or dew. By optimizing surface design and material properties, researchers aim to improve the capture, retention, and removal of water droplets. The integration of MD simulations has become a powerful tool in investigating these systems, providing valuable insights into the mechanisms that govern water harvesting. This section explores various studies that utilize hybrid wetting technologies for water harvesting, offering both experimental and theoretical perspectives on the subject.

Ostwal et al. [111] examined water harvesting from air using conductive polymers, specifically polyaniline (PANI) doped with HCl or HBr. This study utilized MD simulations to model the absorption and diffusion of water vapor in these polymers at various temperatures. The simulation results closely matched experimental adsorption isotherms and demonstrated that this conductive polymer has a high potential for water harvesting from the air. One innovative aspect of this research is the use of the electrical properties of polyaniline to enhance the process of water absorption and release. Due to its unique characteristics such as chemical and mechanical stability and ease of processing, this polymer is proposed as an ideal material for water harvesting. The simulations showed that at higher temperatures, water vapor permeability in polyaniline increases, and the activation energy for this process aligns well with experimental data. Ultimately, the study suggests that this conductive polymer could effectively be used for water harvesting in various temperature conditions, with potential improvements in efficiency through further doping with other halogen acids.

Lyu et al. [112] proposed a simple and scalable method for fabricating hydrophobic surfaces containing hydrophilic groups to optimize water harvesting (Fig. 29). These surfaces were created using a dip-coating method, offering high efficiency in regenerating and removing water droplets. An innovative aspect of this research is the combination of hydrophilic groups with long hydrophobic chains, which increases droplet nucleation density while maintaining water repellency. In contrast to surfaces inspired by desert beetles, which use both hydrophilic and hydrophobic materials, this study utilizes limited hydrophilic regions to capture mist while the hydrophobic background helps remove droplets. These surfaces increased the water harvesting yield to 3.145 g per square centimeter per hour, significantly higher than many other reported materials. The results show that this method is not only effective for water-harvesting surface construction but also scalable for large-scale production, making it a practical and efficient solution for addressing water shortages, especially in arid and desert regions.

Xu et al. [113] investigated energy harvesting from a hydrochloric acid-water solution flow through carbon nanotubes (CNTs) using MD simulations. The main innovation of this study is explaining the molecular mechanisms of energy harvesting through the movement of chloride ions near the walls of carbon nanotubes (Fig. 30). The liquid flow generates voltage in the direction of the flow, resulting from the movement of ions in the layers surrounding the nanotube. This study demonstrates that the voltage generated depends on the flow rate, ambient temperature, solution concentration, and the nanotube diameter. The simulation results highlighted the “stick-slip” motion of chloride ions and their nucleation mechanism in the solution layers, leading to an electric current opposing the direction of the liquid flow. The generated voltage increases with faster flow rates and higher ion concentrations but decreases as the nanotube diameter increases. This research contributes to a deeper understanding of nanofluid behavior and is specifically proposed for designing high-efficiency energy harvesting devices from mechanical and thermal sources. Additionally, the effects of temperature and nanotube diameter on energy harvesting efficiency were demonstrated, aligning with previous experimental trends.

Martinez et al. [114] explored water harvesting from air using

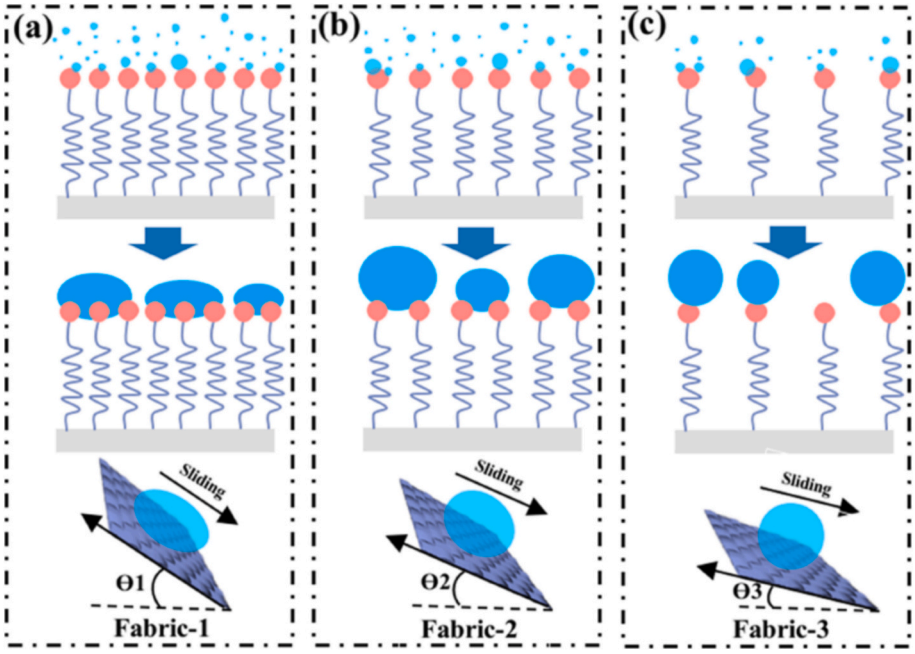

Fig. 29. Schematic of droplet nucleation and removal on (a) fabric-1, (b) fabric-2, and (c) fabric-3 in the work presented by Lyu et al. [112].

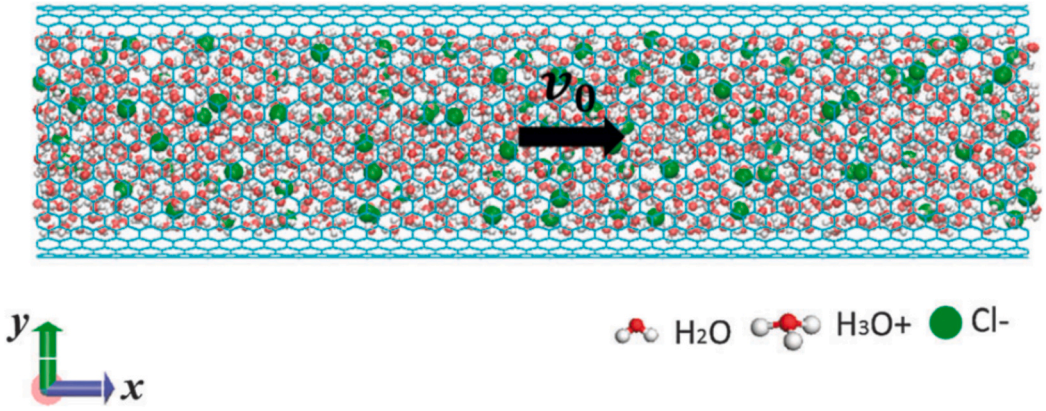

Fig. 30. Flow of hydrochloric acid-water solution through carbon nanotubes in the work presented by Xu et al. [113] (Reproduced from ref. [113] with permission from the PCCP Owner Societies.)

thermoresistant ionic liquids and studied the mechanism of water absorption and nanostructuring of these liquids using MD simulations. The main innovation of this research lies in the use of thermoresistant ionic liquids for separating water from air by leveraging temperature changes to enhance water absorption and release. These ionic liquids can absorb water vapor at low temperatures and release water at high temperatures. The ionic liquid studied, tetrabutylphosphonium 2,4-dimethylbenzenesulfonate, combines with water in a 50–50 ratio at 36 °C. Simulation results showed that water molecules are rapidly absorbed onto the ionic liquid surface, forming three-dimensional water clusters that create nanostructures with the ionic liquid. This process can lead to the formation of sub-surface water cavities, which then combine into nanostructures throughout the sample. The simulations suggest that strong nanostructuring can improve water separation from the ionic liquid even before macroscopic phase separation occurs. This research highlights the potential for using this method for water harvesting in dry and desert conditions and as an effective approach to supplying water using solar energy.

Wang et al. [115] reviewed recent advancements in designing nature-inspired surfaces for water harvesting. The paper focuses on technologies for harvesting water from fog, dew, and humidity, as well as solar evaporation. The main innovation of this research is applying biomimetic principles (nature-inspired design) to develop surfaces with specific wettability properties that enhance water harvesting efficiency. Research shows that living organisms like desert beetles, cacti, and spider webs use unique mechanisms to collect water. The article explores these mechanisms and how they can be applied to the design of synthetic surfaces. For example, microscopic structures on cacti help collect and channel water to the base, while spider webs effectively capture dew. These principles contribute to designing surfaces with wettability gradients and micro- and nanostructures that increase water collection efficiency from fog and dew. The paper also addresses the challenges in developing sustainable water harvesting systems, including reducing energy consumption, increasing surface durability, and optimizing water absorption and release processes. The use of eco-friendly materials and low-cost manufacturing methods is highlighted as key to the success of these technologies.

Leivas and Barbosa [116] studied the water absorption process by functionalized carbon nanocones using MD simulations in three different scenarios: a single nanocone in contact with a liquid water reservoir, a nanocone in contact with water vapor, and multiple nanocones in contact with vapor. The results showed that water flows through the nanocones when their tips are functionalized with hydrophilic groups. In contact with vapor, water droplets only form when hydrophilic groups are present at the base of the nanocones, and water flows more linearly and efficiently than in the liquid state. Additionally, the distance between nanocones is a crucial factor in the efficiency of water harvesting.

Overall, hybrid wetting surfaces provide a promising avenue for improving water harvesting efficiency, particularly in resource-constrained environments. By leveraging both hydrophilic and hydrophobic properties, these surfaces optimize water capture and removal, ensuring higher yields and scalability. The integration of nanostructured surfaces, conductive polymers, and biomimetic designs further enhances these systems' potential to meet water demands in dry regions. Future research should continue to explore the interplay between material properties and environmental conditions to develop more efficient and cost-effective water harvesting solutions.

Water harvesting applications leverage hybrid wettability surfaces to enhance the collection and removal of water droplets. Lyu et al. [112] reported that hybrid surfaces combining hydrophobic and hydrophilic groups achieved water collection yields of 3.145 g/cm^2/h, outperforming traditional superhydrophilic surfaces. However, the durability of these surfaces under real-world environmental conditions—such as exposure to UV radiation, temperature fluctuations, and contaminants—remains a significant limitation. Studies have shown that surface degradation, such as the loss of hydrophobic coatings or erosion of nanostructures, can reduce water collection efficiency by up to 40 % after prolonged use in harsh environments. The integration of more robust materials, such as graphene-based coatings or functionalized carbon nanotubes, has been proposed to mitigate these issues. Additionally, self-healing coatings and bio-inspired designs (e.g., desert beetle-inspired patterns) show promise for maintaining functionality over extended periods. Scaling water harvesting systems for community-level deployment also presents challenges, particularly in optimizing cost and energy requirements for fabrication and regeneration. Developing scalable, cost-effective methods for creating and maintaining these hybrid surfaces is critical for their adoption in arid and semi-arid regions.

5.5. *Oil-water separation*

Oil-water separation is a critical application area where hybrid wetting surfaces have shown tremendous potential. The ability to manipulate surface wettability plays a key role in enhancing the separation efficiency between oil and water phases. In oil-water separation systems, materials that exhibit both hydrophobic and hydrophilic properties can be engineered to selectively allow the passage of one fluid while repelling the other. This dual functionality is highly valuable for a range of industries, including petroleum refining, wastewater treatment, and environmental remediation. Clay minerals, for example, significantly impact oil-water separation through their influence on wettability, a complex property shaped by surface chemistry, fluid composition, and geological conditions. Key minerals such as kaolinite,

montmorillonite, illite, mica, talc, and pyrophyllite have been studied using experimental methods and MD simulations to understand their behavior in multiphase systems [117]. Inconsistencies in wettability data often arise from varying conditions, but techniques like sessile drop and captive bubble contact angle measurements (Fig. 31a), along with dynamic contact angle assessments (Fig. 31b, c), provide crucial insights. This fact has been shown by Pan et al. [117]. Their review helps clarify these discrepancies, offering valuable guidance for optimizing oil-water separation processes.

In the context of hybrid wetting for oil-water separation, surface properties, wettability, and interfacial tension are key factors for optimizing separation processes. Zheng and Bai [118] investigated polymer nanocomposites designed to improve mechanical and structural properties under various temperature and pressure conditions. Their work focused on combining nanoparticles with polymer matrices to enhance strength, elasticity, and thermal resistance (Fig. 32). While their study primarily focused on material properties, their findings on nanoparticle behavior under different environmental conditions offer valuable insights for designing hybrid wetting surfaces with enhanced durability and stability, potentially applicable to oil-water separation systems that operate in harsh environments.

Zhao and Wen [119] demonstrated through pore-scale simulations that altering wettability and reducing interfacial tension in porous media can significantly enhance oil recovery, highlighting the importance of hybrid wetting surfaces in facilitating multiphase flow (Fig. 33). This suggests that precisely controlled surface properties at the pore level are critical for improving oil-water separation.

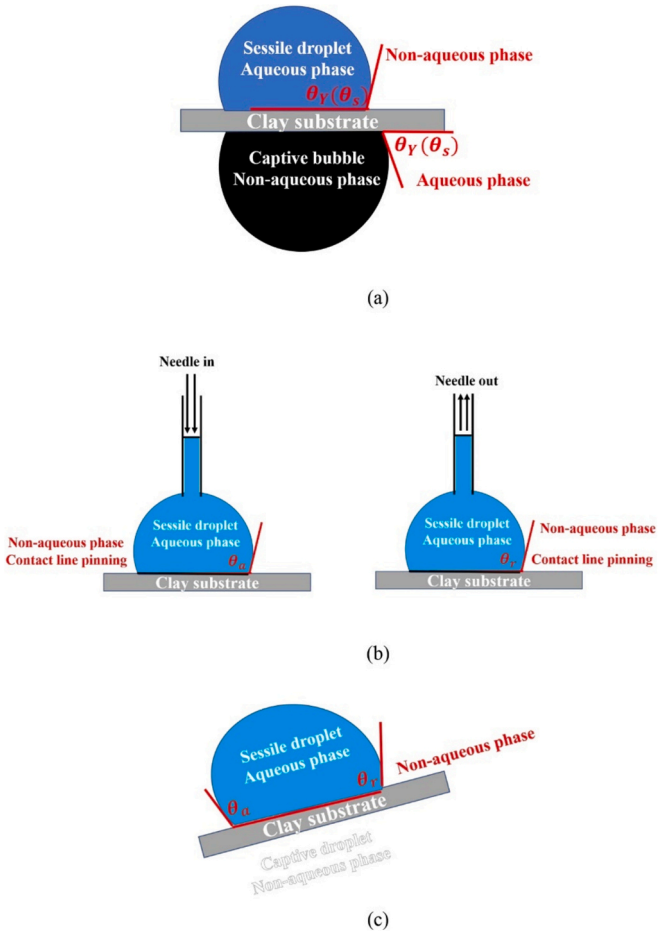

Fig. 31. (a) Diagram showing the sessile drop and captive bubble techniques used for measuring contact angles; (b) and (c) diagrams depicting the process of dynamic contact angle measurements. [117].

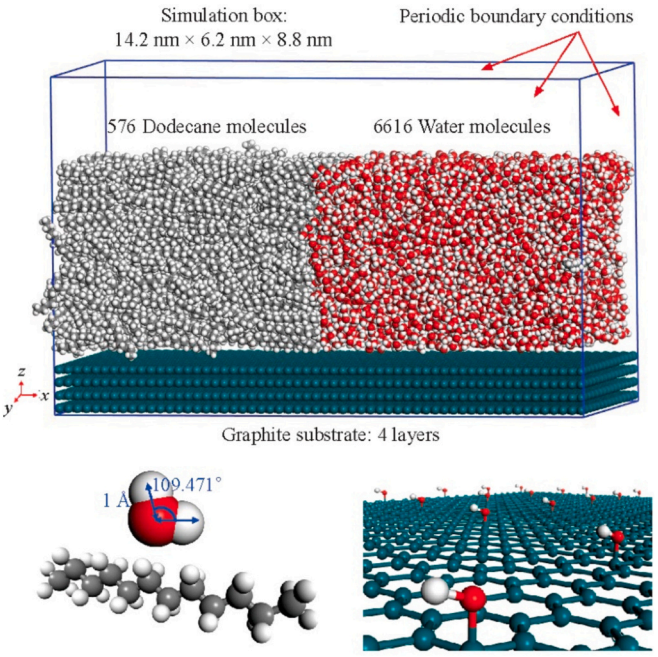

Fig. 32. Oil-water-graphite system in Zheng and Bai's work [118].

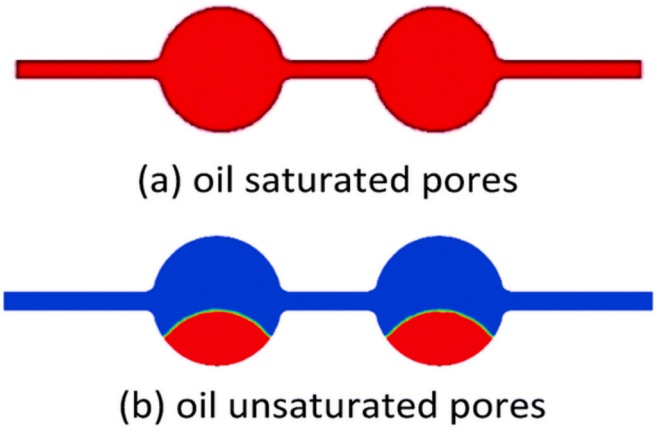

Fig. 33. Initial distribution of oil/water phases for oil-saturated and unsaturated pores in the work by Zhao and Wen [119].

Zahedi and Foroutan [120] used MD simulations to investigate oil-water separation on polymethyl methacrylate surfaces enhanced with titanium dioxide (TiO_2) nanoparticles. Their study showed that the superhydrophilic properties of TiO_2 nanoparticles effectively promote water clustering, enhancing separation efficiency (Fig. 34). This underscores the potential of nanoparticles in hybrid wetting systems to facilitate selective liquid interaction and improve oil-water separation.

Foroutan et al. [121] focused on using multilayer graphene oxide membranes for oil-water separation, finding that reducing interlayer spacing and pore size selectively allows water to pass while restricting oil flow (Fig. 35). The hydrophilic functional groups on the graphene oxide membranes play a crucial role, providing a highly selective separation process driven by hybrid wetting principles.

Li et al. [122] developed a superhydrophobic/superoleophilic surface using SiO_2-TiO_2 nanoparticles, achieving over 99 % separation efficiency for oil-water mixtures and emulsions. The high surface roughness and low surface energy were shown to enhance separation, with MD simulations supporting these findings by demonstrating stable oil interactions on the surface.

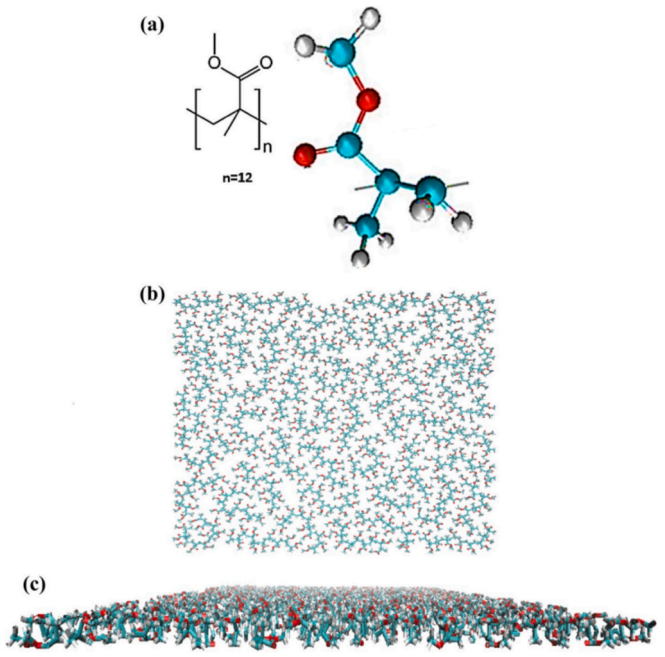

Fig. 34. (a) Molecular structure of a methyl methacrylate monomer unit: cyan (carbon), white (hydrogen), and red (oxygen) atoms, (b) Top view of a flat polymethyl methacrylate sheet, and (c) Side view of the polymethyl methacrylate sheet [120]. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

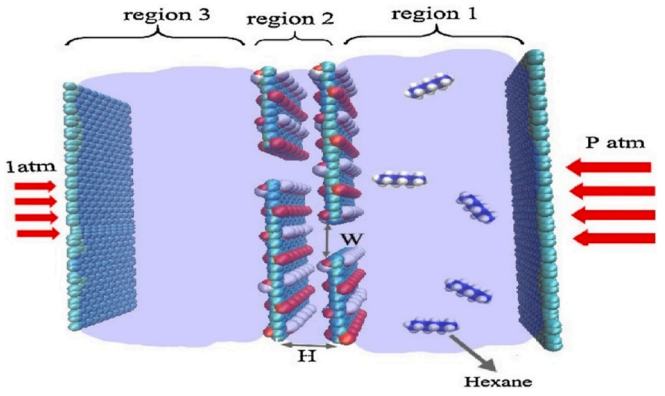

Fig. 35. Schematic representation of oil-water separation using graphene oxide membranes in the study by Foroutan et al. [121].

Xiong et al. [123] examined nanoscale pore wettability in shale formations, where water uptake was found to be faster than oil absorption in hydrophilic pores. Their results emphasize the critical balance between hydrophilic and hydrophobic properties in hybrid wetting surfaces for controlling fluid flow in porous materials, making them highly relevant to oil-water separation processes.

Ma et al. [124] explored the impact of ionic liquids on interfacial tension and quartz wettability in oil-water systems. They found that ionic liquids could significantly reduce interfacial tension and improve oil droplet mobility on hydrophilic surfaces, indicating their potential in enhancing separation efficiency in conjunction with hybrid wetting technologies.

To summarize, hybrid wetting surfaces offer a versatile and efficient approach to tackling the challenges associated with oil-water separation. By precisely controlling surface wettability and incorporating advanced materials, it is possible to enhance separation performance in both traditional and novel applications. The research findings presented here highlight the diverse strategies, from nanoparticle enhancements to membrane technologies, that leverage hybrid wetting principles to achieve superior results. As the field advances, continued exploration of material innovations and surface engineering will further optimize these systems, making them even more effective in complex and demanding environments.

5.6. Other applications

In addition to the well-established applications of hybrid wetting in areas such as thermal management and heat transfer, microfluidics and lab-on-a-chip systems, self-cleaning surfaces, water harvesting, and oil-water separation (as shown in the previous sections), hybrid wetting continues to demonstrate its versatility in various other fields. The ability to precisely control the interaction between liquids and surfaces with mixed hydrophilic and hydrophobic regions has opened new avenues for innovation across multiple industries. These applications extend into areas such as dynamic wetting transitions, enhanced material properties, and advanced boiling processes, where hybrid wetting plays a crucial role in optimizing performance. Studies exploring these emerging areas have highlighted the impact of surface design, nanostructuring, and energy interactions, underscoring the importance of hybrid wetting in addressing complex engineering and scientific challenges. The following section will review key advancements in these other applications of hybrid wetting, further illustrating its broad potential.

Khan and Singh [125] utilized MD simulations to explore the wetting transition of nanoscale water droplets on textured surfaces. Their innovative analysis focused on the transition between Wenzel and Cassie-Baxter states, which depended on pillar height and surface coverage. The results showed that as pillar height increases, the wetting behavior shifts from the Wenzel state (where droplets penetrate the surface texture) to the Cassie state (where air is trapped beneath the droplet). This transition is influenced by droplet size and the system's scale, with shorter pillars showing more significant contact angle fluctuations. Free energy calculations revealed that the Wenzel state is more stable due to lower free energy. The study also introduced the phantom-wall method as an effective tool for calculating free energy in different wetting states, offering insights for designing superhydrophobic surfaces for water-repellent, anti-icing, and anti-adhesion applications.

Ghani Dehkordi [126] investigated the impact of electric field intensity and microchannel material (platinum and gold) on the boiling dynamics of an H_2O/Fe_3O_4 nanofluid using MD simulations. The study revealed that an increased electric field accelerates the boiling process, raising the velocity, temperature, and density of the nanofluid. Peak values reached 0.0071 Å per picosecond, 714 Kelvin, and 0.038 atoms per cubic angstrom. Additionally, phase transitions occurred more quickly in gold microchannels compared to platinum ones, demonstrating the role of electric field strength and channel material in optimizing nanofluid boiling. The system modeled, including the Fe_3O_4 nanoparticles within the nanofluid, is shown schematically in Fig. 36.

Ali et al. [127] explored the influence of nanoparticles on the thermal and dynamic properties of polymeric materials through MD simulations and thermodynamic analysis. Their focus was on how nanoparticle size, type, and distribution affect the mechanical and thermal properties of polymers. Results showed that incorporating nanoparticles significantly enhanced the mechanical strength, thermal stability, and conductivity of the polymer matrix. Smaller nanoparticles with larger surface areas had the most pronounced effect on mechanical property improvement. Additionally, variations in nanoparticle distribution influenced heat dissipation, with potential applications in industries such as automotive, aerospace, and electronics.

Zhang et al. [128] conducted a novel investigation into the surface wettability of coal at different stages of coalification, combining experimental methods with MD simulations. The study covered low-rank coal, bituminous coal, and anthracite coal, demonstrating that

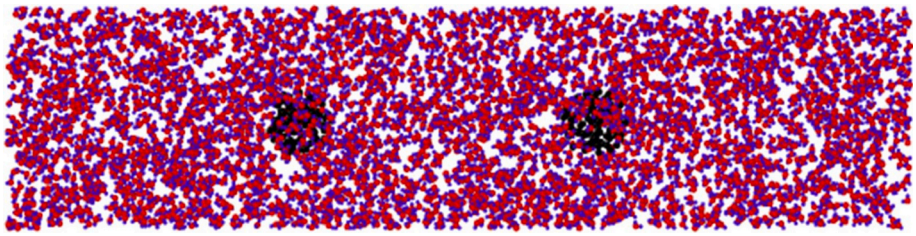

Fig. 36. Molecular simulation of the H_2O/Fe_3O_4 nanofluid with two Fe3O4 nanoparticles (shown in black) [126].

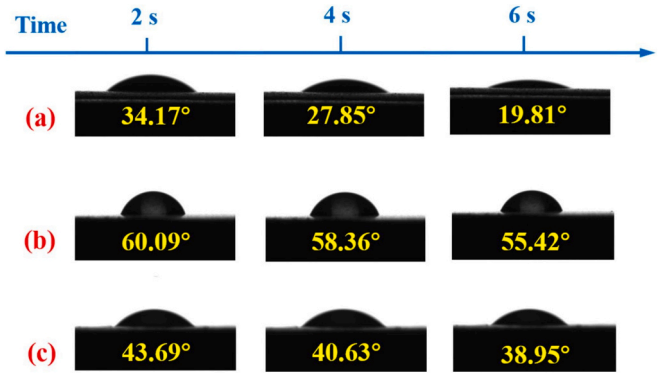

Fig. 37. Water contact angle of (a) low-rank coal, (b) bituminous coal, and (c) anthracite coal [128].

water contact angles on coal surfaces increased with coalification before decreasing. Bituminous coal exhibited the highest hydrophobicity, while low-rank coal showed the lowest, attributed to the abundance of oxygen-containing hydrophilic groups on low-rank coal surfaces. Fig. 37 shows the water contact angles for low-rank coal, bituminous coal, and anthracite coal, further illustrating these trends. MD simulations revealed that the molecular structure and number of hydrogen bonds between water molecules and the coal surface were critical in determining wettability, with low-rank coal forming the most hydrogen bonds. This study provides valuable insights into optimizing the coal flotation process.

These studies collectively highlight the diverse applications of hybrid wetting in enhancing material properties, optimizing fluid behavior, and improving energy efficiency across different fields.

6. Geometrical parameters of Nanochannels on condensation

The size and shape of channels, as key geometric parameters, play an important role in the condensation process. Condensation is one of the key processes in heat transfer, utilized in various industrial systems such as heat exchangers, cooling systems, and distillation equipment. The geometric characteristics of channels, particularly at micro and nano scales, directly influence condensation behavior and the thermal efficiency of the system. One of the most important geometric parameters affecting condensation is the channel size. Smaller channels tend to increase the rate of heat transfer due to the higher surface-to-volume ratio. This occurs because, in smaller channels, the contact surface between the vapor and the channel walls increases, leading to more efficient condensation. Studies have shown that reducing the diameter of channels can enhance the condensation rate and improve the thermal performance of the system. In addition to size, the shape of the channel significantly impacts condensation behavior. Different cross-sectional shapes, such as circular, rectangular, and elliptical, can influence the flow pattern and the accumulation of condensation droplets [129,130].

One of the primary challenges in designing channels with different sizes and shapes is managing the flow of liquid and vapor. In smaller and more complex cross-sections, the likelihood of vapor flow blockage and reduced condensation efficiency increases. Therefore, the optimal design of channel size and shape should both enhance the condensation rate and prevent flow blockage. This is particularly important in microchannels and nanochannels, where precise flow control is crucial for optimal performance [131].

In nanochannels, the effect of channel size and shape on condensation behavior becomes more pronounced due to surface phenomena such as van der Waals forces and surface tension. These forces can significantly alter the behavior of liquid and vapor at the nanoscale, potentially leading to unpredictable changes in condensation rates. The shape of the channel can also influence the stability of condensation droplets and how they are removed from the surface. In channels with asymmetric cross-sections, such as elliptical or trapezoidal, different shear forces act on the droplets, which can result in faster droplet removal and reduced thermal resistance. This characteristic can improve the efficiency of cooling systems and heat exchangers [132]. Another impact of channel size and shape on condensation is the control of temperature distribution along the channel. In smaller and more complex cross-sections, the temperature distribution can be highly heterogeneous, potentially leading to the formation of hot and cold spots along the channel. This thermal heterogeneity can destabilize the condensation process and reduce the overall system efficiency. Therefore, optimizing the shape and size of channels to achieve uniform temperature distribution is of great importance [133,134].

To better understand the role of geometrical parameters, such as channel size and shape, in influencing the condensation process at both micro- and nanoscale levels, an in-depth review of relevant studies was conducted. These parameters are critical because they directly affect heat transfer rates and the overall efficiency of condensation, which is particularly important in industrial applications requiring precise control of thermal dynamics. The findings from these studies, which focus on various channel designs, material properties, and condensation conditions, are summarized in Table 2. This table provides a structured comparison of research focuses, methodologies, and key findings, facilitating a comprehensive understanding of how geometrical variations in nanochannels impact both condensation behavior and heat transfer performance.

Most of the mentioned studies emphasize that optimizing surface geometry and utilizing materials with specific properties can significantly improve heat transfer and condensation performance. Amphiphilic surfaces and nanostructures embedded in the fields of distillation and boiling are among those highlighted. Some studies focus on the management and control of boundary phenomena such as condensation and boiling at nano and microscale levels, which underscores the complexity and importance of these processes in heat transfer. Analyses indicate that variations in temperature, pressure, and material properties can dramatically affect the behavior and performance of condensation and boiling systems, making these factors crucial in material selection and system design. Several studies address challenges related to material strength and durability under harsh operating conditions, emphasizing the importance of selecting suitable materials for specific applications.

Precisely determining the effects of scale on physical and chemical

Table 2
Summary of studies on geometrical parameters of nanochannels and their effect on condensation.

Study	Research Focus	Key Findings
Alwazzan et al. [134]	Condensation heat transfer performance in copper tubes with hybrid hydrophobic and less hydrophobic patterns.	Hybrid patterns showed significant improvement in condensation heat transfer compared to filmwise condensation.
Cui et al. [135]	Effect of Cu nanoparticles on thermal behavior and phase transition of Ar atomic fluid using MD simulations.	Cu nanoparticles improved heat flux and thermal conductivity of Ar fluid.
Ganapathy et al. [136]	Transient numerical model for heat transfer during condensation in microchannels using VOF method.	Predicted flow patterns matched experimental data, with Â ± 7.5 % for pressure drop and Â ± 15.6 % for Nusselt number.
Wang et al. [137]	Heterogeneous condensation of methane gas using MD simulations.	High-energy walls and lower cold source temperature increased methane condensation and nucleation.
Wang et al. [138]	Heat transfer characteristics of condensation in nanochannels with nanopillars using MD simulations.	Nanopillars enhanced condensation performance, with increased nanopillar height improving heat transfer by 11.6 % and 35.8 %.
Ghahremanian et al. [139]	Annular condensation of Ar vapor in nanochannels with focus on flow and heat transfer.	Nanostructure height improved condensation, with significant effects on wall temperature.
Abadi et al. [140]	Effect of slope angles on vapor condensation in compact and elongated channels using VOF method.	Increased slope angle and vapor mass, with decreased saturation temperature, improved heat transfer coefficient.
Yang et al. [141]	Capillary condensation of water in nanostructured channels using van der Waals aggregation.	Kelvin equation is valid in highly hydrophilic channels; water condensation influenced by humidity variations.
Hu et al. [142]	Condensation heat transfer in finned tubes with combined hydrophilic and hydrophobic surfaces in presence of non-condensable gas.	Finned tube with combined hydrophilic-superhydrophobic surface exhibited highest heat transfer performance with non-condensable gas.
Egab et al. [143]	Enhancement of condensation heat transfer rates using circular wettability contrast patterns on copper surfaces.	Optimization of circular patterns significantly improved condensation heat transfer.
Hekmatifar et al. [144]	Behavior of nanofluids and their phase change on surfaces with varying roughness using MD simulations.	Rough surfaces and copper nanoparticles improved condensation and phase change rates.
Liao and duan [145]	Factors influencing surface condensation on nanostructured surfaces with angular and size variations using MD.	Taller nanostructures accelerated condensation on hydrophilic surfaces and reduced condensation on hydrophobic surfaces.
Peng et al. [87]	Impact of spherical roughness on Ar flow in microchannels under boiling conditions using MD simulations.	Spherical roughness had negligible effect on Ar flow behavior at lower external forces.
Ghahremanian et al. [146]	Effects of nanostructured surface features on condensation and heat transfer in nanochannels using MD.	Rough surfaces enhanced particle distribution and heat transfer in nanochannels during condensation.
Shah [147]	Development of a correlation for heat transfer during condensation in horizontal micro and mini channels.	New correlation improved prediction accuracy for heat transfer rates in micro and mini channels.
Sun et al. [148]	Hybrid MD-continuum simulation for gas flow distillation in a microchannel.	Developed hybrid model provided accurate simulation of gas flow distillation in microchannels.
Welch and Piri [149]	Reverse condensation in oil-coated nanochannels using MD simulations.	Reverse condensation observed in oil-coated nanochannels at higher pressures.
Winter and McCarthy [150]	Self-draining process of droplets on amphiphilic surfaces in miniaturized channels under condensation conditions.	Amphiphilic surfaces improved droplet self-draining and reduced thermal resistance.
Zhang et al. [151]	Membrane distillation in mid-scale graphene channels using coarse-grained MD models.	Diffusion flux in graphene channels was higher than typical microporous membranes.
Zhong et al. [152]	Distillation process in dead-end nanochannels shorter than 100 nm.	Vapor flow resistance limited condensate growth in nanochannels shorter than 100 nm.
Zhong et al. [153]	Capillary condensation in 8 nm depth channels.	Condensation starts earlier in smaller channels, driven by liquid-vapor interfaces.

properties, especially at the nanoscale, is one of the main challenges. These limitations can impact the accuracy of both theoretical and experimental predictions. Establishing repeatable experimental conditions and clear standards for evaluating and comparing results across different studies remains a challenge. The complexity of modeling thermodynamic and flow systems with high accuracy is also a challenge, particularly at small scales where quantum and surface effects dominate. Many of the materials and processes used in research may have adverse environmental impacts, requiring careful review and management. Additionally, the new technologies proposed in many papers require more time and investment to reach industrial standards. Systems that operate in harsh environments need materials with high resistance to temperature and chemical changes.

7. Challenges and future directions

7.1. Challenges

- Research in the field of nanostructured surfaces and phenomena related to wetting and condensation continues to face numerous challenges that require further investigation in the future. One of the main challenges in this area is the complexity of modeling at the nanoscale. Molecular dynamics (MD) simulations, due to computational limitations and insufficient accuracy in simulating molecular forces, are unable to precisely predict the behavior of droplet condensation at very small scales. To overcome these limitations, the development of multiscale simulation methods that can model condensation phenomena simultaneously at both nano and macro scales is essential.
- Another challenge lies in the instability of nanostructured surfaces under variable environmental conditions, such as changes in temperature and humidity. Although these surfaces perform well in laboratory settings, their performance becomes unstable in industrial conditions where temperature and humidity fluctuate suddenly. Enhancing the durability and stability of these surfaces in challenging environments is a critical area for future research. Additionally, the high cost of producing nanostructured surfaces with optimized roughness and precise patterns remains a significant barrier to the widespread commercialization of this technology. Developing cheaper and scalable production methods for industrial applications will require further research and development.

- A key challenge in using nanostructured surfaces with combined wettability properties is their unstable performance under changing environmental conditions like temperature and humidity. In industrial environments, severe fluctuations in these conditions can affect surface performance. Therefore, future research should focus on improving the stability and durability of these surfaces.
- Producing surfaces with combined hydrophilic and hydrophobic properties and optimal roughness is expensive, which is one of the main challenges in commercializing this technology. Developing cheaper and more scalable production methods for industrial applications is essential.
- Molecular dynamics simulations also face limitations due to the lack of valid and repeatable experimental data. Reliable experimental data is crucial to verify the results of simulations and theoretical models, which remains a fundamental challenge in this field.
- MD simulations provide valuable insights into condensation and hydrophilicity phenomena, but due to their nanoscale nature, there is a significant gap in the experimental validation of these simulations. Existing experimental techniques often lack the precision or resolution required to validate nanoscale predictions.
- As proposed solutions, advanced imaging techniques such as Atomic Force Microscopy (AFM) and Environmental Scanning Electron Microscopy (ESEM) can directly observe nucleation, growth, and droplet movement on nanostructured surfaces. These techniques provide real-time data that can be used to validate simulation results.
- Additionally, the development of microfluidic platforms with integrated sensors can simulate nanoscale environments, enabling experimental validation of MD simulations under controlled conditions. Combining MD simulations with continuum-scale methods like Computational Fluid Dynamics (CFD) can further bridge the gap between nanoscale predictions and experimental observations. Using experimental data as boundary conditions for MD simulations can also enhance model accuracy.

7.2. Future research recommendations

- It is recommended to conduct more studies on condensation using molecular dynamics in the fields of nanomedicine and biotechnology. Precise control over wetting and the design of nanostructured channels could lead to the development of new technologies in nanomedicine and targeted drug delivery. Hybrid-wettability nanostructured channels could be used in microfluidic systems to control biological and chemical flows.
- One of the most important areas for molecular dynamics studies is investigating condensation on nanostructured surfaces with combined wettability. These surfaces can be utilized in thermal power plants to increase heat transfer efficiency and reduce energy consumption. Optimizing wettability in nanoscale heat transfer systems will help prevent channel blockages and increase fluid flow stability.
- Molecular dynamics studies should focus on exploring the behavior of nanostructured surfaces with combined wettability in various industrial sectors, particularly the chemical industry. Designing hybrid surfaces with optimized roughness can help reduce operational costs in distillation and purification processes.
- Future research could focus on improving the accuracy and speed of molecular dynamics simulations to study condensation behavior at the nanoscale. These studies could provide better results for industrial applications and contribute to optimizing different systems.

8. Conclusion

The studies presented in this report investigate wetting and condensation phenomena on nanostructured surfaces using molecular dynamics simulations, highlighting the importance of these concepts in improving the performance of heat transfer and condensation systems. The results indicate that nanostructured surfaces with combined wettability properties (hydrophilic and hydrophobic) can significantly increase heat transfer rates and contribute to the stability of dropwise condensation. These hybrid surfaces enable the optimization of condensation systems' performance and reduce thermal resistance. Additionally, optimizing geometry and controlling surface roughness are factors that can improve the efficiency of these processes and reduce energy costs in industrial applications.

However, challenges such as precise control over droplet behavior and the stability of nanostructured surfaces under variable environmental conditions remain, requiring further research. This research emphasizes the importance of accurate modeling and the use of multi-scale technologies to address these challenges, showing that combining theoretical and experimental data can enhance the performance of heat transfer systems in industries such as renewable energy, microelectronics, and medical technologies.

CRediT authorship contribution statement

Adnan I. Khdair: Writing – original draft, Conceptualization. **Saeed Aghakhani:** Writing – original draft. **N.H. Thi:** Writing – review & editing. **Masoud Afrand:** Writing – review & editing, Supervision, Project administration, Conceptualization.

Declaration of competing interest

None.

References

[1] J. De la Rosa, A. Escriva, L. Herranz, T. Cicero, J. Munoz-Cobo, Review on condensation on the containment structures, Prog. Nucl. Energy 51 (2009) 32–66.
[2] A.S. Dalkilic, S. Wongwises, Intensive literature review of condensation inside smooth and enhanced tubes, Int. J. Heat Mass Transf. 52 (2009) 3409–3426.
[3] N.S. Singh, J. Zhang, J. Stafford, C. Anthony, N. Gao, Implementing superhydrophobic surfaces within various condensation environments: a review, Adv. Mater. Interfaces 8 (2021) 2001442.
[4] A. Goswami, S.C. Pillai, G. McGranaghan, Surface modifications to enhance dropwise condensation, Surf. Interf. 25 (2021) 101143.
[5] S. Adera, L. Naworski, A. Davitt, N.K. Mandsberg, A.V. Shneidman, J. Alvarenga, J. Aizenberg, Enhanced condensation heat transfer using porous silica inverse opal coatings on copper tubes, Sci. Rep. 11 (2021) 10675.
[6] P.S. Mahapatra, A. Ghosh, R. Ganguly, C.M. Megaridis, Key design and operating parameters for enhancing dropwise condensation through wettability patterning, Int. J. Heat Mass Transf. 92 (2016) 877–883.
[7] M. Edalatpour, L. Liu, A. Jacobi, K. Eid, A. Sommers, Managing water on heat transfer surfaces: a critical review of techniques to modify surface wettability for applications with condensation or evaporation, Appl. Energy 222 (2018) 967–992.
[8] N. Miljkovic, D.J. Preston, E.N. Wang, Recent developments in altered wettability for enhancing condensation, in: Encyclopedia of Two-Phase Heat Transfer and Flow II: Special Topics and Applications Chapter 4, 2016, pp. 85–131.
[9] R. Gulfam, Y. Chen, Recent growth of wettability gradient surfaces: a review, Research 2022 (2022) 9873075.
[10] S.-F. Zheng, U. Gross, X.-D. Wang, Dropwise condensation: from fundamentals of wetting, nucleation, and droplet mobility to performance improvement by advanced functional surfaces, Adv. Colloid Interf. Sci. 295 (2021) 102503.
[11] Z. Li, J. Zhong, D.A. Levin, B.J. Garrison, Development of homogeneous water condensation models using molecular dynamics, AIAA J. 47 (2009) 1241–1251.
[12] D. Toghraie, M. Hekmatifar, Y. Salehipour, M. Afrand, Molecular dynamics simulation of Couette and Poiseuille water-copper nanofluid flows in rough and smooth nanochannels with different roughness configurations, Chem. Phys. 527 (2019) 110505.
[13] A.H. Meghdadi Isfahani, I. Tasdighi, A. Karimipour, E. Shirani, M. Afrand, A joint lattice Boltzmann and molecular dynamics investigation for thermohydraulical simulation of nano flows through porous media, Europ. J. Mechan. B/Fluids 55 (2016) 15–23.
[14] Z. Liang, A. Chandra, E. Bird, P. Keblinski, A molecular dynamics study of transient evaporation and condensation, Int. J. Heat Mass Transf. 149 (2020) 119152.
[15] D. Toghraie, M. Mokhtari, M. Afrand, Molecular dynamic simulation of copper and platinum nanoparticles Poiseuille flow in a nanochannels, Phys. E. 84 (2016) 152–161.
[16] D.T. Ranathunga, A. Shamir, X. Dai, S.O. Nielsen, Molecular dynamics simulations of water condensation on surfaces with tunable wettability, Langmuir 36 (2020) 7383–7391.
[17] Y. Peng, M. Zarringhalam, M. Hajian, D. Toghraie, S.J. Tadi, M. Afrand, Empowering the boiling condition of argon flow inside a rectangular

microchannel with suspending silver nanoparticles by using of molecular dynamics simulation, J. Mol. Liq. 295 (2019) 111721.
[18] S. AlRawashdeh, K.H. Barakat, Applications of molecular dynamics simulations in drug discovery, in: M. Gore, U.B. Jagtap (Eds.), Computational Drug Discovery and Design, Springer US, New York, NY, 2024, pp. 127–141.
[19] I. Tanasawa, Recent advances in condensation heat transfer, in: International Heat Transfer Conference Digital Library, Begel House Inc., 1994.
[20] S.A. Hollingsworth, R.O. Dror, Molecular dynamics simulation for all, Neuron 99 (2018) 1129–1143.
[21] B. Leimkuhler, C. Matthews, Molecular dynamics, Interdiscip. Appl. Math. 39 (2015).
[22] A.U. Khan, M. Khan, M.H. Cho, M.M. Khan, Selected nanotechnologies and nanostructures for drug delivery, nanomedicine and cure, Bioprocess Biosyst. Eng. 43 (2020) 1339–1357.
[23] G. Thakral, R. Thakral, N. Sharma, J. Seth, P. Vashisht, Nanosurface–the future of implants, J. Clin. Diagn. Res. 8 (2014) ZE07.
[24] R.M. Streicher, M. Schmidt, S. Fiorito, Nanosurfaces and nanostructures for artificial orthopedic implants, Nanomedicine 2 (2007) 861–874.
[25] M. Rosoff, Nano-Surface Chemistry, Marcel Dekker, New York, 2002.
[26] A.R. Cherian, L. Benny, A. George, A. Varghese, G. Hegde, Recent advances in functionalization of carbon nanosurface structures for electrochemical sensing applications: tuning and turning, J. Nanostructure Chem. (2021) 1–26.
[27] K.K. Chenab, R. Eivazzadeh-Keihan, P. Pashazadeh-Panahi, A. Mokhtarzadeh, A. Maleki, Biological applications of bacterial Nano-surface layers: a brief overview, Nanomed. J. 5 (2018).
[28] N. Sirdeshmukh, G. Dongre, Laser micro & nano surface texturing for enhancing osseointegration and antimicrobial effect of biomaterials: a review, Mater. Today Proc. 44 (2021) 2348–2355.
[29] C. Li, J. Yang, W. He, M. Xiong, X. Niu, X. Li, D.G. Yu, A review on fabrication and application of tunable hybrid micro–nano array surfaces, Adv. Mater. Interfaces 10 (2023) 2202160.
[30] J. Morel, Low Temperature Hydrothermal Titanium Oxide Nanosurfaces for Functional Applications, UNSW Sydney, 2024.
[31] K. Komatsu, T. Matsuura, J. Cheng, D. Kido, W. Park, T. Ogawa, Nanofeatured surfaces in dental implants: contemporary insights and impending challenges, Int. J. Implant Dentistry 10 (2024) 34.
[32] E.M. Akinoglu, A.J. Morfa, M. Giersig, Nanosphere lithography-exploiting self-assembly on the nanoscale for sophisticated nanostructure fabrication, Turk. J. Phys. 38 (2014) 563–572.
[33] W.G. Anderson, Wettability literature survey-Part 4: effects of wettability on capillary pressure, J. Pet. Technol. 39 (1987) 1283–1300.
[34] W. Anderson, Wettability literature survey-Part 2: wettability measurement, J. Pet. Technol. 38 (1986) 1246–1262.
[35] S. Ma, X. Zhang, N. Morrow, X. Zhou, Characterization of wettability from spontaneous imbibition measurements, J. Can. Pet. Technol. 38 (1999).
[36] Y. Guo, Z. Guo, W. Liu, Bionic multifunctional fibrous materials for efficient oil/water separation, Droplet 2 (2023).
[37] L. Susana, F. Campaci, A.C. Santomaso, Wettability of mineral and metallic powders: applicability and limitations of sessile drop method and Washburn's technique, Powder Technol. 226 (2012) 68–77.
[38] S. Ebnesajjad, Surface tension and its measurement, in: Handbook of Adhesives and Surface Preparation, Elsevier, 2011, pp. 21–30.
[39] K. Grundke, K. Pöschel, A. Synytska, R. Frenzel, A. Drechsler, M. Nitschke, A. Cordeiro, P. Uhlmann, P. Welzel, Experimental studies of contact angle hysteresis phenomena on polymer surfaces—toward the understanding and control of wettability for different applications, Adv. Colloid Interf. Sci. 222 (2015) 350–376.
[40] D.D. Davis, R. Horton, J.L. Heitman, T. Ren, Wettability and hysteresis effects on water sorption in relatively dry soil, Soil Sci. Soc. Am. J. 73 (2009) 1947–1951.
[41] M. Shoji, X.Z. Yi, Study of contact angle hysteresis: in relation to boiling surface wettability, JSME Int. J. Ser. B Fluids Therm. Eng. 37 (1994) 560–567.
[42] P. Ustohal, F. Stauffer, T. Dracos, Measurement and modeling of hydraulic characteristics of unsaturated porous media with mixed wettability, J. Contam. Hydrol. 33 (1998) 5–37.
[43] P. Zanganeh, A.S. Goharrizi, S. Ayatollahi, M. Feilizadeh, H. Dashti, Efficiency improvement of solar stills through wettability alteration of the condensation surface: an experimental study, Appl. Energy 268 (2020) 114923.
[44] M.M. Taheri, B. Rezaee, H. Pakzad, A. Moosavi, Condensation heat transfer enhancement through durable, self-propelling fluorine-free silane-treated anodized surfaces, J. Mater. Chem. A 12 (40) (2024) 27327–27339.
[45] A. Ghosh, S. Beaini, B.J. Zhang, R. Ganguly, C.M. Megaridis, Enhancing dropwise condensation through bioinspired wettability patterning, Langmuir 30 (2014) 13103–13115.
[46] C.R. Kharangate, I. Mudawar, Review of computational studies on boiling and condensation, Int. J. Heat Mass Transf. 108 (2017) 1164–1196.
[47] D. Bhuyan, Direct and Indirect Contact Filmwise as well as Dropwise Condensation of Water Vapour with and without Noncondensible Gas-A Review, Int. Res. J. Eng. Technol 7 (10) (2020) 415–419.
[48] S. Qin, R. Ji, C. Miao, L. Jin, C. Yang, X. Meng, Review of enhancing boiling and condensation heat transfer: surface modification, Renew. Sust. Energ. Rev. 189 (2024) 113882.
[49] M.T. Hughes, S. Garimella, A review of active enhancement methods for boiling and condensation, Int. J. Heat Mass Transf. 218 (2024) 124752.
[50] R. Gao, M. Song, C.Y.H. Chao, S. Lin, L. Zhang, X. Zhang, Review on condensation frosting and defrosting experiments for superhydrophobic surfaces, Appl. Therm. Eng. 236 (2024) 121691.
[51] A. Kumar, R. Kumar, A.K. Das, An inclusive review on structural enhancement techniques in forced condensation inside tubes, Int. J. Heat Fluid Flow 106 (2024) 109297.
[52] T. Josyula, L. Kumar Malla, T.M. Thomas, S.S. Kalichetty, P. Sinha Mahapatra, A. Pattamatta, Fundamentals and applications of surface wetting, Langmuir 40 (2024) 8293–8326.
[53] S. Afkhami, Challenges of numerical simulation of dynamic wetting phenomena: a review, Curr. Opin. Colloid Interface Sci. 57 (2022) 101523.
[54] R. Blossey, Self-cleaning surfaces—virtual realities, Nat. Mater. 2 (2003) 301–306.
[55] L. Hou, X. Liu, X. Ge, R. Hu, Z. Cui, N. Wang, Y. Zhao, Designing of anisotropic gradient surfaces for directional liquid transport: fundamentals, construction, and applications, The Innovation 4 (6) (2023) 100508.
[56] S. Paul, M.N. Hasan, Molecular dynamics perspective of condensation over a hybrid wetting surface, Mol. Simul. 48 (2022) 448–461.
[57] J. Zhang, M.K. Borg, J.M. Reese, Multiscale simulation of dynamic wetting, Int. J. Heat Mass Transf. 115 (2017) 886–896.
[58] G. Lu, X.-D. Wang, Y.-Y. Duan, A critical review of dynamic wetting by complex fluids: from Newtonian fluids to non-Newtonian fluids and nanofluids, Adv. Colloid Interf. Sci. 236 (2016) 43–62.
[59] P. Ge, S. Wang, J. Zhang, B. Yang, Micro−/nanostructures meet anisotropic wetting: from preparation methods to applications, Mater. Horiz. 7 (2020) 2566–2595.
[60] E. Wagemann, Y. Wang, S. Das, S.K. Mitra, Wettability of nanostructured hexagonal boron nitride surfaces: molecular dynamics insights on the effect of wetting anisotropy, Phys. Chem. Chem. Phys. 22 (2020) 2488–2497.
[61] D. Quéré, Wetting and Roughness 38, 2008, pp. 71–99.
[62] S. Gao, W. Liu, Z. Liu, Tuning nanostructured surfaces with hybrid wettability areas to enhance condensation, Nanoscale 11 (2019) 459–466.
[63] S. Paul, M. Nasim Hasan, Effects of gradient and patterned wetting configuration on condensation over hybrid wetting surface from atomistic viewpoint, J. Mol. Liq. 359 (2022) 119347.
[64] S. Paul, D. Chakraborty, S.J. Esha, M.N. Hasan, Role of wettability contrast on nanoscale condensation over hybrid wetting surface with gradient and patterned wetting configuration at various philic-phobic content, Surf. Interf. 36 (2023) 102417.
[65] X. Zhang, F. Shi, J. Niu, Y. Jiang, Z. Wang, Superhydrophobic surfaces: from structural control to functional application, J. Mater. Chem. 18 (2008) 621–633.
[66] A. Lafuma, D. Quéré, Superhydrophobic states, Nat. Mater. 2 (2003) 457–460.
[67] N.A. Patankar, Mimicking the Lotus effect: influence of double roughness structures and slender pillars, Langmuir 20 (2004) 8209–8213.
[68] A.-K. Hossain, M.N. Miah, M.N. Hasan, A molecular dynamics study on thin film phase change characteristics over nano-porous surfaces with hybrid wetting conditions, Int. Commun. Heat Mass Transf. 156 (2024) 107599.
[69] M. Zhou, L. Fu, F. Jiang, B. Jiang, D. Drummer, Atomistic investigation on the wetting behavior and interfacial joining of polymer-metal, Interface 12 (2020) 1696.
[70] M. Rezaei, D.M. Warsinger, J.H. Lienhard, M.C. Duke, T. Matsuura, W. M. Samhaber, Wetting phenomena in membrane distillation: mechanisms, reversal, and prevention, Water Res. 139 (2018) 329–352.
[71] Y. Zhang, P.C. Millett, M. Tonks, L. Zhang, B. Biner, Molecular dynamics simulations of He bubble nucleation at grain boundaries, J. Phys. Condens. Matter 24 (2012) 305005.
[72] Z. Ye, S. Li, C. Wang, R. Shen, W. Wen, Capillary flow control in nanochannels via hybrid surface, RSC Adv. 6 (2016) 2774–2777.
[73] X. Zhang, H. Liu, L. Jiang, Wettability and applications of nanochannels, Adv. Mater. 31 (2019) 1804508.
[74] T. Zhang, F. Javadpour, X. Li, K. Wu, J. Li, Y. Yin, Mesoscopic method to study water flow in nanochannels with different wettability, Phys. Rev. E 102 (2020) 013306.
[75] A. De La Escosura-Muñiz, A. Merkoçi, Nanochannels preparation and application in biosensing, ACS Nano 6 (2012) 7556–7583.
[76] Z. Ye, S. Li, B. Zhou, Y.S. Hui, R. Shen, W. Wen, Nanofluidic mixing via hybrid surface, Appl. Phys. Lett. 105 (2014).
[77] Y. Dong, Z. Zhao, J. Zhao, Z. Guo, G. Du, Y. Sun, D. He, J. Duan, J. Liu, H. Yao, High-performance osmotic power generators based on the 1D/2D hybrid nanochannel system, ACS Appl. Mater. Interfaces 14 (2022) 29197–29212.
[78] R. Fallahzadeh, F. Bozzoli, L. Cattani, N. Naeimabadi, A comprehensive review on molecular dynamics simulations of forced convective heat transfer in Nanochannels, Energies 17 (2024) 4352.
[79] Z. Zhang, L. Wen, L. Jiang, Bioinspired smart asymmetric nanochannel membranes, Chem. Soc. Rev. 47 (2018) 322–356.
[80] W. Sparreboom, A. van den Berg, J.C. Eijkel, Transport in nanofluidic systems: a review of theory and applications, New J. Phys. 12 (2010) 015004.
[81] X. Chen, L. Zhang, Review in manufacturing methods of nanochannels of bio-nanofluidic chips, Sensors Actuators B Chem. 254 (2018) 648–659.
[82] H. Daiguji, Ion transport in nanofluidic channels, Chem. Soc. Rev. 39 (2010) 901–911.
[83] H. Zhao, L. Zhou, X. Du, Bubble nucleation on grooved surfaces with hybrid wettability: molecular dynamics study under a transient temperature boundary condition, Int. J. Heat Mass Transf. 166 (2021) 120752.
[84] K.G. Dehkordi, A. Karimipour, M. Afrand, D. Toghraie, A.H.M. Isfahani, Molecular dynamics simulation concerning nanofluid boiling phenomenon affected by the external electric field: effects of number of nanoparticles through Pt, Fe, and Au microchannels, J. Mol. Liq. 324 (2021) 114775.

[85] G. Liang, Y. Chen, J. Wang, Z. Wang, S. Shen, Experiments and modeling of boiling heat transfer on hybrid-wettability surfaces, Int. J. Multiphase Flow 144 (2021) 103810.
[86] Q. Cao, W. Shao, X. Ren, X. Ma, K. Shao, Z. Cui, Y. Liu, Molecular dynamics simulations of the liquid film evaporation heat transfer on different wettability hybrid surfaces at the nanoscale, J. Mol. Liq. 314 (2020) 113610.
[87] Y. Peng, M. Zarringhalam, A.A. Barzinjy, D. Toghraie, M. Afrand, Effects of surface roughness with the spherical shape on the fluid flow of argon atoms flowing into the microchannel, under boiling condition using molecular dynamic simulation, J. Mol. Liq. 297 (2020) 111650.
[88] S. Cai, Q. Li, W. Li, L. Zhang, X. Liu, Effects of mole fraction and surface wettability on evaporation of Ar/Kr mixtures: a molecular dynamics study, J. Mol. Liq. 319 (2020) 114189.
[89] D. Holder, A. Peter, M. Kirsch, S. Cáceres, R. Weber, V. Onuseit, R. Kulenovic, J. Starflinger, T. Graf, Enhancing heat transfer at low temperatures by laser functionalization of the inner surface of metal pipes, Sci. Rep. 14 (2024) 2557.
[90] W. Deng, S. Ma, W. Li, H. Liu, J. Zhao, A molecular dynamics investigation of boiling heat transfer over wettability thermo-responsive surface, Int. J. Heat Mass Transf. 191 (2022) 122856.
[91] W. Guo, L. Zeng, Y. Lu, J. Chen, Tuning surface hybrid-wettability to enhance the vapour film phenomenon induced by boiling heat transfer: molecular dynamics, Int. Commun. Heat Mass Transf. 136 (2022) 106172.
[92] L. Zhang, J. Xu, Nucleate boiling of thin liquid films on nanostructured surfaces with hybrid wettability using molecular dynamics simulation, J. Mol. Liq. 366 (2022) 120272.
[93] C. Li, B. Kim, J. Yoon, S. Sett, J. Oh, Advances in directional wetting surfaces for enhanced fluid control: a comprehensive review, Adv. Funct. Mater. 34 (2024) 2308265.
[94] M. Hiratsuka, M. Emoto, A. Konno, S. Ito, Molecular dynamics simulation of the influence of nanoscale structure on water wetting and condensation, Micromachines 10 (2019).
[95] B. Xu, Z. Chen, Molecular dynamics study of water vapor condensation on a composite wedge-shaped surface with multi wettability gradients, Int. Commun. Heat Mass Transf. 105 (2019) 65–72.
[96] T. Li, M. Li, L. Zhang, M. Yan, H. Li, Molecular dynamics study of the temperature-dependent coalescence of liquid Nanodrops: implications for microfluidics, ACS Appl. Nano Mater. 2 (2019) 7978–7988.
[97] J. Sun, Y.L. He, W.Q. Tao, J.W. Rose, H.S. Wang, Multi-scale study of liquid flow in micro/nanochannels: effects of surface wettability and topology, Microfluid. Nanofluid. 12 (2012) 991–1008.
[98] H.-B. Kwon, S.-J. Yoo, Y.-J. Kim, Microfluidic condensation nanoparticle counter using water as the condensing liquid for assessing individual exposure to airborne nanoparticles, Lab Chip 20 (2020) 1092–1102.
[99] X. Chen, M. Wang, Y. Xin, Y. Huang, One-step fabrication of self-cleaning superhydrophobic surfaces: a combined experimental and molecular dynamics study, Surf. Interf. 31 (2022) 102022.
[100] W. Qiang, Z. Lan, Y. Liu, Y. Liu, W. Xu, R. Wen, X. Ma, Excellent hydrophobicity and high mobility of condensate droplets on hierarchical nanostructured surfaces: insights from MD simulations, J. Mol. Liq. 407 (2024) 125139.
[101] H. Ren, S. Zhu, Y. Xiao, C. Li, Self-propelled Nanodroplet jumping enhanced by Nanocone arrays: implications for self-cleaning and anti-icing surfaces, ACS Appl. Nano Mater. 5 (2022) 810–817.
[102] X. Hao, Y. Yang, S. Dong, H. Zheng, R. Wang, Robust superhydrophobic self-cleaning coating prepared by silane modified multi-walled carbon nanotubes: a combined experimental and molecular dynamics study, J. Mol. Graph. Model. 132 (2024) 108831.
[103] K. Zhang, Z. Li, M. Maxey, S. Chen, G.E. Karniadakis, Self-cleaning of hydrophobic rough surfaces by coalescence-induced wetting transition, Langmuir 35 (2019) 2431–2442.
[104] J.H. Pu, S.K. Wang, J. Sun, W. Wang, H.S. Wang, Growth and self-jumping of single condensed droplet on nanostructured surfaces: a molecular dynamics simulation, J. Mol. Liq. 340 (2021) 116902.
[105] W. Ding, M. Wang, X. Dai, J. Zhang, G. Xin, X. Wang, Dewetting transition of water on nanostructured and wettability patterned surfaces: a molecular dynamics study, J. Mol. Liq. 336 (2021) 116869.
[106] K.M. Wisdom, J.A. Watson, X. Qu, F. Liu, G.S. Watson, C.-H. Chen, Self-cleaning of superhydrophobic surfaces by self-propelled jumping condensate 110 (2013) 7992–7997.
[107] P. Wang, X. Sun, H. Lv, S. Ma, Z. Wang, Investigation of surface wettability and their influencing mechanisms under vibration field: a molecular dynamics simulation study, Comput. Mater. Sci. 197 (2021) 110615.
[108] D.L. Miller, A Comparison between Self-Cleaning Properites Via Rolling Droplets and Condensation on Superhydrophobic Surfaces, Brigham Young University, 2017.
[109] Y.-N. Dong, N.-N. Han, X.-J. Zhang, B.-X. Zhang, J.-Y. Wang, X. He, Fluid transportation by droplets impacting wettability-controlled surfaces at the nanoscale: a molecular dynamics simulation study, Microfluid. Nanofluid. 26 (2022) 101.
[110] J. Song, R. Shi, X. Bai, H. Algadi, D. Sridhar, An overview of surface with controllable wettability for microfluidic system, intelligent cleaning, water harvesting, and surface protection, Adv. Compos. Hybrid Mater. 6 (2022) 22.
[111] M.M. Ostwal, M. Sahimi, T.T. Tsotsis, Water harvesting using a conducting polymer: a study by molecular dynamics simulation, Phys. Rev. E 79 (2009) 061801.
[112] P. Lyu, X. Zhang, X. Jiang, B. Shang, X. Liu, Z. Deng, One-step preparation of hydrophobic surfaces containing hydrophilic groups for efficient water harvesting, Langmuir 37 (2021) 9630–9636.
[113] B. Xu, X. Chen, Liquid flow-induced energy harvesting in carbon nanotubes: a molecular dynamics study, Phys. Chem. Chem. Phys. 15 (2013) 1164–1168.
[114] N.C. Forero-Martinez, R. Cortes-Huerto, L. Ward, P. Ballone, Water harvesting by Thermoresponsive ionic liquids: a molecular dynamics study of the water absorption kinetics and of the role of Nanostructuring, J. Phys. Chem. B 127 (2023) 5494–5508.
[115] Y. Wang, W. Zhao, M. Han, J. Xu, K.C. Tam, Biomimetic surface engineering for sustainable water harvesting systems, Nat. Water 1 (2023) 587–601.
[116] R. Leivas, M.C. Barbosa, Functionalized carbon nanocones performance in water harvesting, J. Chem. Phys. 158 (2023) 194702.
[117] B. Pan, X. Yin, S. Iglauer, A review on clay wettability: from experimental investigations to molecular dynamics simulations, Adv. Colloid Interf. Sci. 285 (2020) 102266.
[118] W. Zheng, C. Sun, B. Bai, Molecular dynamics study on the effect of surface hydroxyl groups on three-phase wettability in oil-water-graphite systems, Polymers 9 (2017).
[119] J. Zhao, D.J.R.A. Wen, Pore-scale simulation of wettability and interfacial tension effects on flooding process for enhanced oil recovery 7 (2017) 41391–41398.
[120] H. Zahedi, M. Foroutan, Separation of water–oil mixture on poly methyl methacrylate surface using TiO2 nanoparticles via molecular dynamics simulation, Adsorption 25 (2019) 1019–1031.
[121] M. Foroutan, H. Zahedi, E. Soleimani, Investigation of water-oil separation via graphene oxide membranes: a molecular dynamics study, Colloids Surf. A Physicochem. Eng. Asp. 555 (2018) 201–208.
[122] C. Li, Y. Xiong, L. Zhao, H. Wan, J. Li, S. Fang, M. Wang, M. Duan, J. Ren, Y. Xiao, Investigation of Oil–Water Separation on an F-SiO2/TiO2-Based Superhydrophobic/Superoleophilic Surface: Experiment Evaluation and MD Simulation, Langmuir 39 (2023) 1694–1708.
[123] J. Xiong, J. Tang, X. Zhou, X. Liu, L. Liang, L. Hou, Molecular dynamics simulation and experimental studies of the wettability behaviors of shales, Energy Fuel 36 (2022) 3526–3538.
[124] Q. Ma, W. Zhu, Z. Song, J. Zhang, B. Li, W. Bu, B. Pan, Influences and mechanisms of imidazolium-based ionic liquids on oil–water interfacial tension and quartz wettability: experiment and molecular dynamics simulations, Fuel 352 (2023) 129053.
[125] S. Khan, J.K. Singh, Wetting transition of nanodroplets of water on textured surfaces: a molecular dynamics study, Mol. Simul. 40 (2014) 458–468.
[126] K. Ghani Dehkordi, A. Karimipour, M. Afrand, D. Toghraie, A.H. Meghdadi Isfahani, The electric field and microchannel type effects on H2O/Fe3O4 Nanofluid boiling process: molecular dynamics study, Int. J. Thermophys. 41 (2020) 132.
[127] M.S. Ali, J. Cui, N. Ullah, M. Ali Kiani, H. Mei, X. Ren, K. Xu, Z. Fan, X. Mei, Molecular dynamics simulation on wetting of silver Nanosolder on a diamond surface, J. Phys. Chem. C 126 (2022) 7221–7229.
[128] R. Zhang, Y. Xing, Y. Xia, J. Luo, J. Tan, G. Rong, X. Gui, New insight into surface wetting of coal with varying coalification degree: an experimental and molecular dynamics simulation study, Appl. Surf. Sci. 511 (2020) 145610.
[129] H. Wang, J.W. Rose, Film condensation in horizontal microchannels: effect of channel shape, Int. Conf. Nanochannels Microchannels Minichannels 41855 (2005) 729–735.
[130] S. Nebuloni, J.R. Thome, Numerical modeling of laminar annular film condensation for different channel shapes, Int. J. Heat Mass Transf. 53 (2010) 2615–2627.
[131] M.M. Derby, A. Chatterjee, Y. Peles, M.K. Jensen, Flow condensation heat transfer enhancement in a mini-channel with hydrophobic and hydrophilic patterns, Int. J. Heat Mass Transf. 68 (2014) 151–160.
[132] P. Toninelli, S. Bortolin, M. Azzolin, D. Del Col, Effects of geometry and fluid properties during condensation in minichannels: experiments and simulations, Heat Mass Transf. 55 (2019) 41–57.
[133] A. Asadollahi, S. Rashidi, J. Esfahani, Condensation process and phase-change in the presence of obstacles inside a minichannel, Meccanica 52 (2017) 2265–2274.
[134] M. Alwazzan, K. Egab, B. Peng, J. Khan, C. Li, Condensation on hybrid-patterned copper tubes (I): characterization of condensation heat transfer, Int. J. Heat Mass Transf. 112 (2017) 991–1004.
[135] H. Cui, S. Saleem, J.E. Jam, M.H. Beni, M. Hekmatifar, D. Toghraie, R. Sabetvand, Effects of roughness and radius of nanoparticles on the condensation of nanofluid structures with molecular dynamics simulation: statistical approach, J. Taiwan Inst. Chem. Eng. 128 (2021) 346–353.
[136] H. Ganapathy, A. Shooshtari, K. Choo, S. Dessiatoun, M. Alshehhi, M.M. Ohadi, Numerical analysis of condensation of R134a in a single microchannel, in: ASME 2012 International Mechanical Engineering Congress and Exposition, Vol. Volume 7: Fluids and Heat Transfer, Parts A, B, C, and D, 2012, pp. 2971–2980.
[137] Y. Wang, Z. Wang, B. Wang, J. Bian, Y. Hua, W. Cai, Heterogeneous nucleation condensation of methane gas on the wall-a molecular dynamics study, Energy 283 (2023) 129087.
[138] M. Wang, H. Sun, L. Cheng, Flow condensation heat transfer characteristics of Nanochannels with Nanopillars: a molecular dynamics study, Langmuir 37 (2021) 14744–14752.
[139] S. Ghahremanian, A. Abbassi, Z. Mansoori, D. Toghraie, Molecular dynamics simulation of annular condensation of vapor argon through a nanochannel for different saturation conditions with focusing on the flow and heat transfer, Int. Commun. Heat Mass Transf. 116 (2020) 104704.

[140] S.M.A.N.R. Abadi, W.A. Davies, P. Hrnjak, J.P. Meyer, Numerical study of steam condensation inside a long inclined flattened channel, Int. J. Heat Mass Transf. 134 (2019) 450–467.
[141] Q. Yang, P.Z. Sun, L. Fumagalli, Y.V. Stebunov, S.J. Haigh, Z.W. Zhou, I. V. Grigorieva, F.C. Wang, A.K. Geim, Capillary condensation under atomic-scale confinement, Nature 588 (2020) 250–253.
[142] H.W. Hu, G.H. Tang, D. Niu, Experimental investigation of condensation heat transfer on hybrid wettability finned tube with large amount of noncondensable gas, Int. J. Heat Mass Transf. 85 (2015) 513–523.
[143] K. Egab, M. Alwazzan, B. Peng, S.K. Oudah, Z. Guo, X. Dai, J. Khan, C. Li, Enhancing filmwise and dropwise condensation using a hybrid wettability contrast mechanism: circular patterns, Int. J. Heat Mass Transf. 154 (2020) 119640.
[144] M. Hekmatifar, D. Toghraie, B. Mehmandoust, F. Aghadavoudi, S. Ali Eftekhari, Molecular dynamics simulation of condensation phenomenon of nanofluid on different roughness surfaces in the presence of hydrophilic and hydrophobic structures, J. Mol. Liq. 334 (2021) 116036.
[145] M.-J. Liao, L.-Q. Duan, Dependencies of surface condensation on the wettability and nanostructure size, Differences 10 (2020) 1831.
[146] S. Ghahremanian, A. Abbassi, Z. Mansoori, D. Toghraie, Effect of nanostructured surface configuration on the interface properties and heat transfer of condensation process of argon inside nanochannels using molecular dynamics simulation, J. Mol. Liq. 339 (2021) 117281.
[147] M.M. Shah, A correlation for heat transfer during condensation in horizontal mini/micro channels, Int. J. Refrig. 64 (2016) 187–202.
[148] J. Sun, Y.-L. He, W.-Q. Tao, Molecular dynamics–continuum hybrid simulation for condensation of gas flow in a microchannel, Microfluid. Nanofluid. 7 (2009) 407–422.
[149] W.R.W. Welch, M. Piri, Molecular dynamics simulations of retrograde condensation in narrow oil-wet nanopores, J. Phys. Chem. C 119 (2015) 10040–10047.
[150] R.L. Winter, M. McCarthy, Dewetting from amphiphilic Minichannel surfaces during condensation, ACS Appl. Mater. Interfaces 12 (2020) 7815–7825.
[151] H. Zhang, B. Liu, H.T. Kieu, M.S. Wu, K. Zhou, A.W.-K. Law, Coarse-grained molecular dynamics study of membrane distillation through meso-size graphene channels, J. Membr. Sci. 558 (2018) 34–44.
[152] J. Zhong, S.H. Zandavi, H. Li, B. Bao, A.H. Persad, F. Mostowfi, D. Sinton, Condensation in one-dimensional dead-end nanochannels, ACS Nano 11 (2017) 304–313.
[153] J. Zhong, J. Riordon, S.H. Zandavi, Y. Xu, A.H. Persad, F. Mostowfi, D. Sinton, Capillary condensation in 8 nm deep channels, J. Phys. Chem. Lett. 9 (2018) 497–503.